MAS.S60
AI Agents for Cognitive Augmentation

Agent Evaluation & Environments

Week 4

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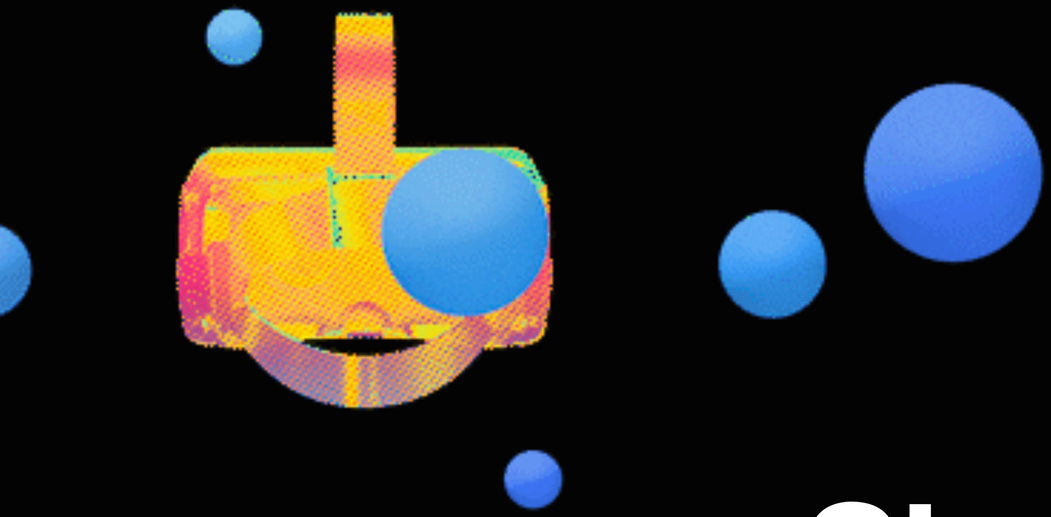
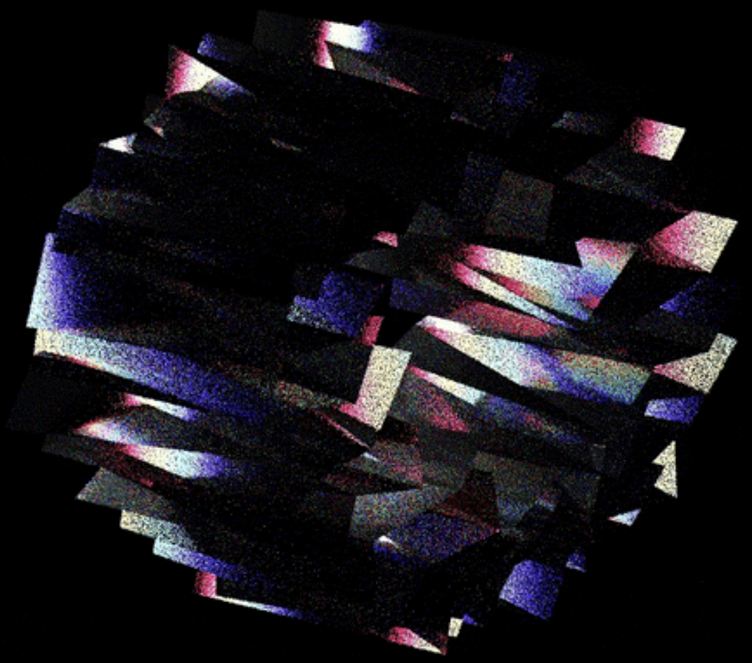
PhD Student
Fluid Interfaces Group
MIT Media Lab

Credit: NASA

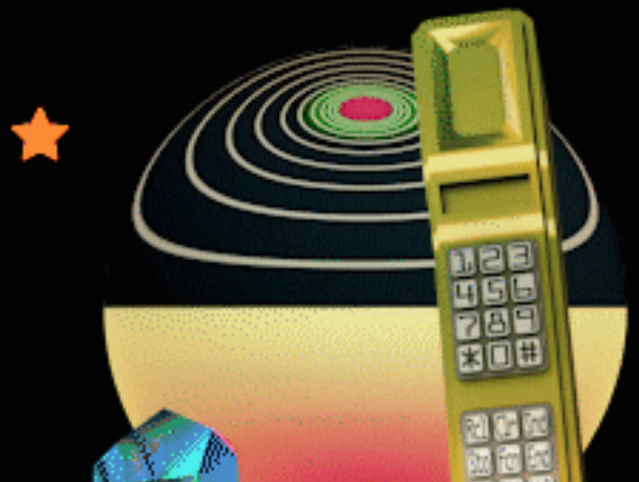
Course logistics

1. **Lab sessions / office hours**
2. **Weekly builds TA feedback**
3. **“Progress review” on the course site**





Show and tell



Show and tell

Zoom Meeting ID: 983 0928 0366

1. **The problem**
   Motivation, agent objective and success criteria.

2. **The solution**
   Diagram of what agent does: (1) environment and agent states, (2) action space, (3) observation space, (4) decision policy (LLM, policy, rules, planner), (4) the reward or criteria telling it whether it's doing well.

3. **Live demo**

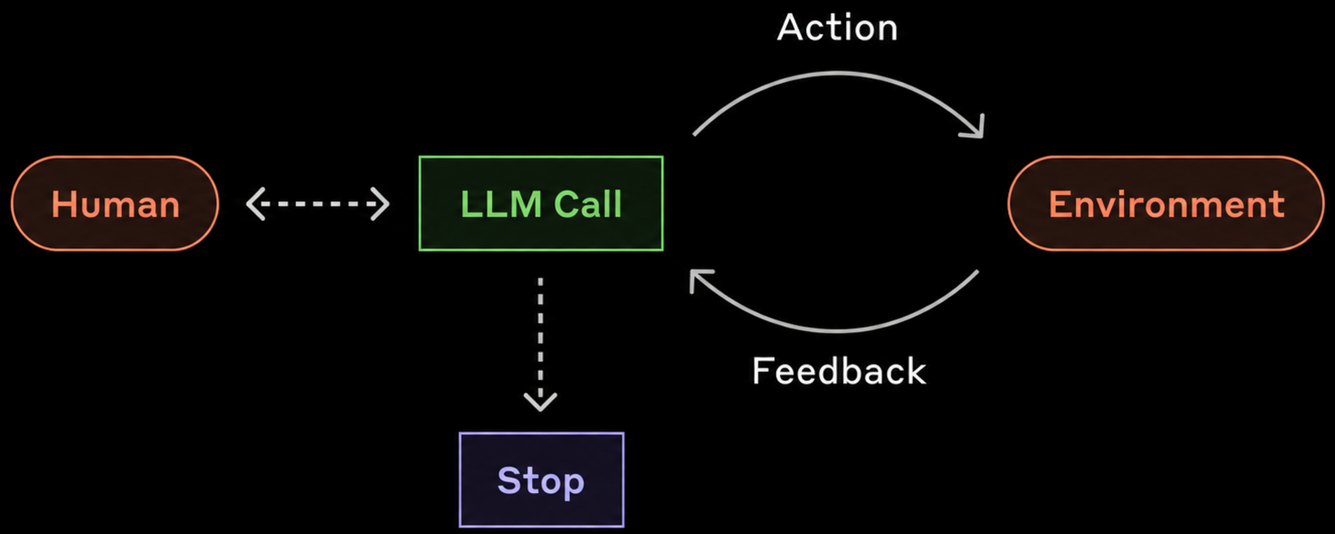
4. **What I learned**
   What went well, one surprise, failure mode, or limitation.

ENVIRONMENTS

Credit: NASA

What is an environment?

What is an environment?



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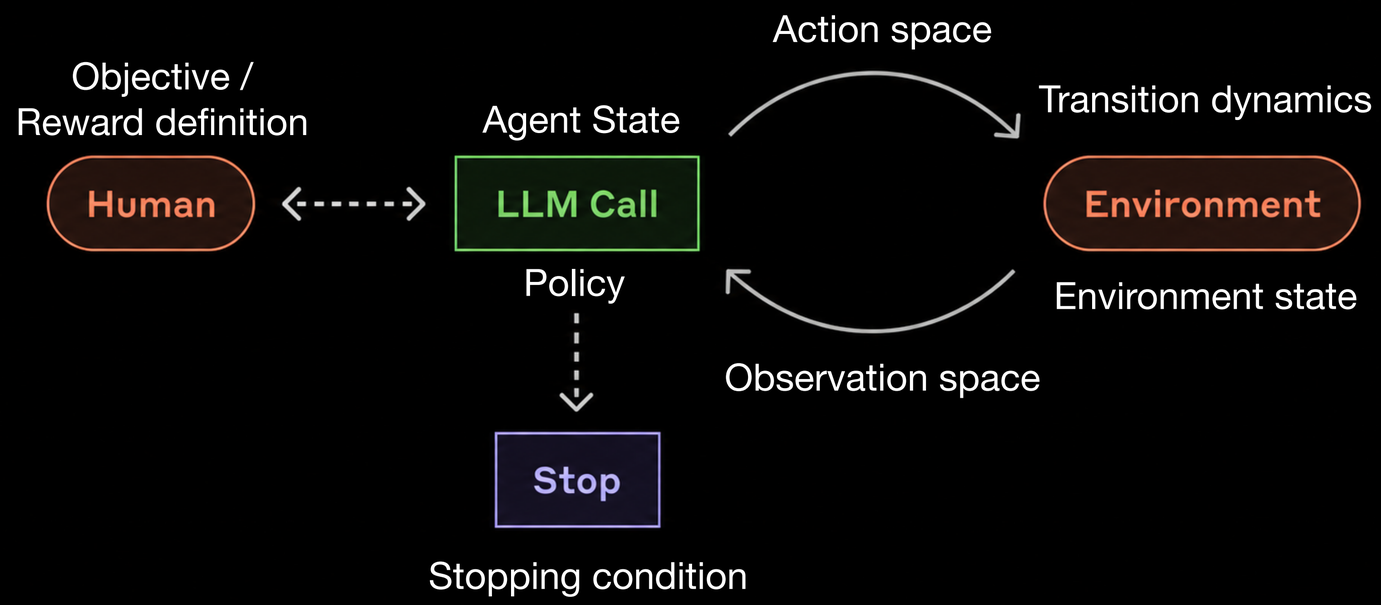
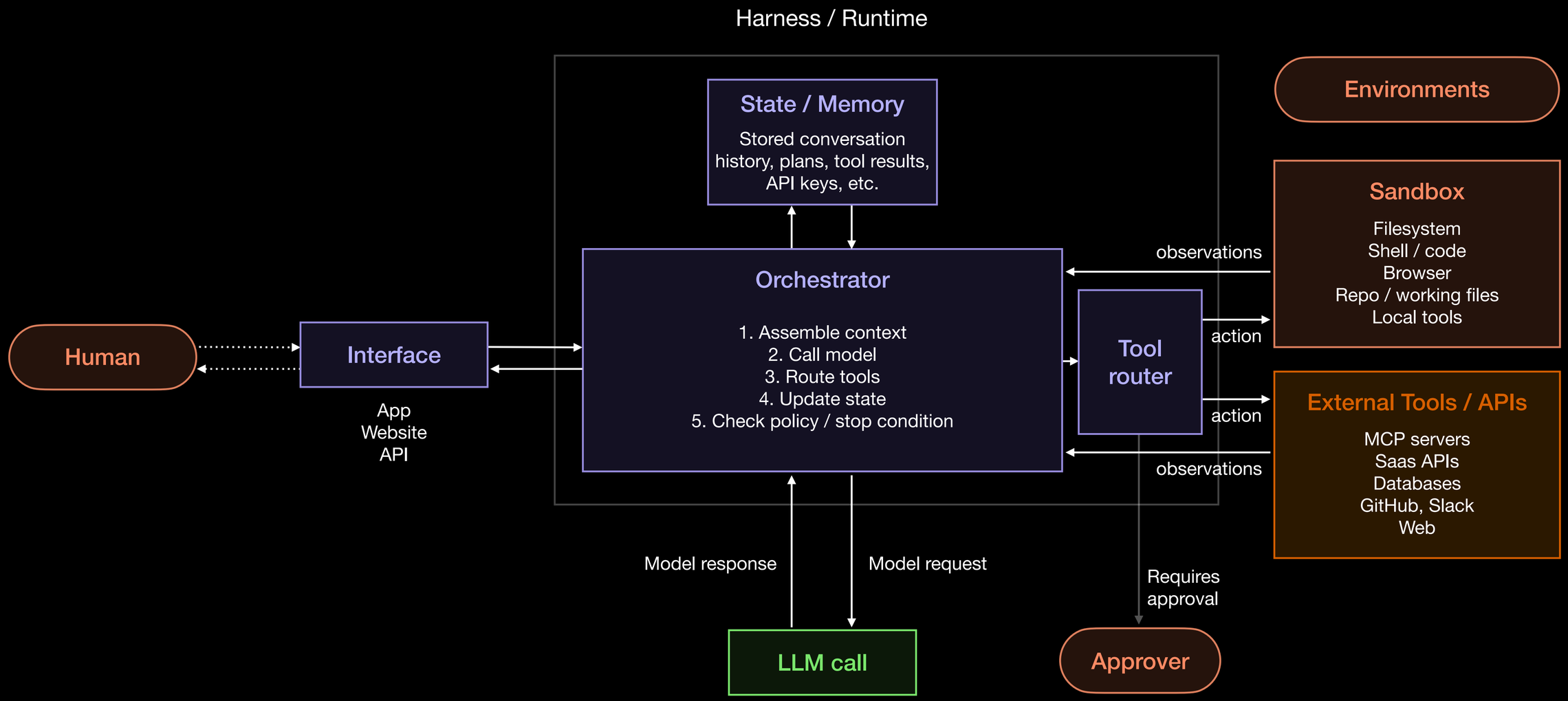
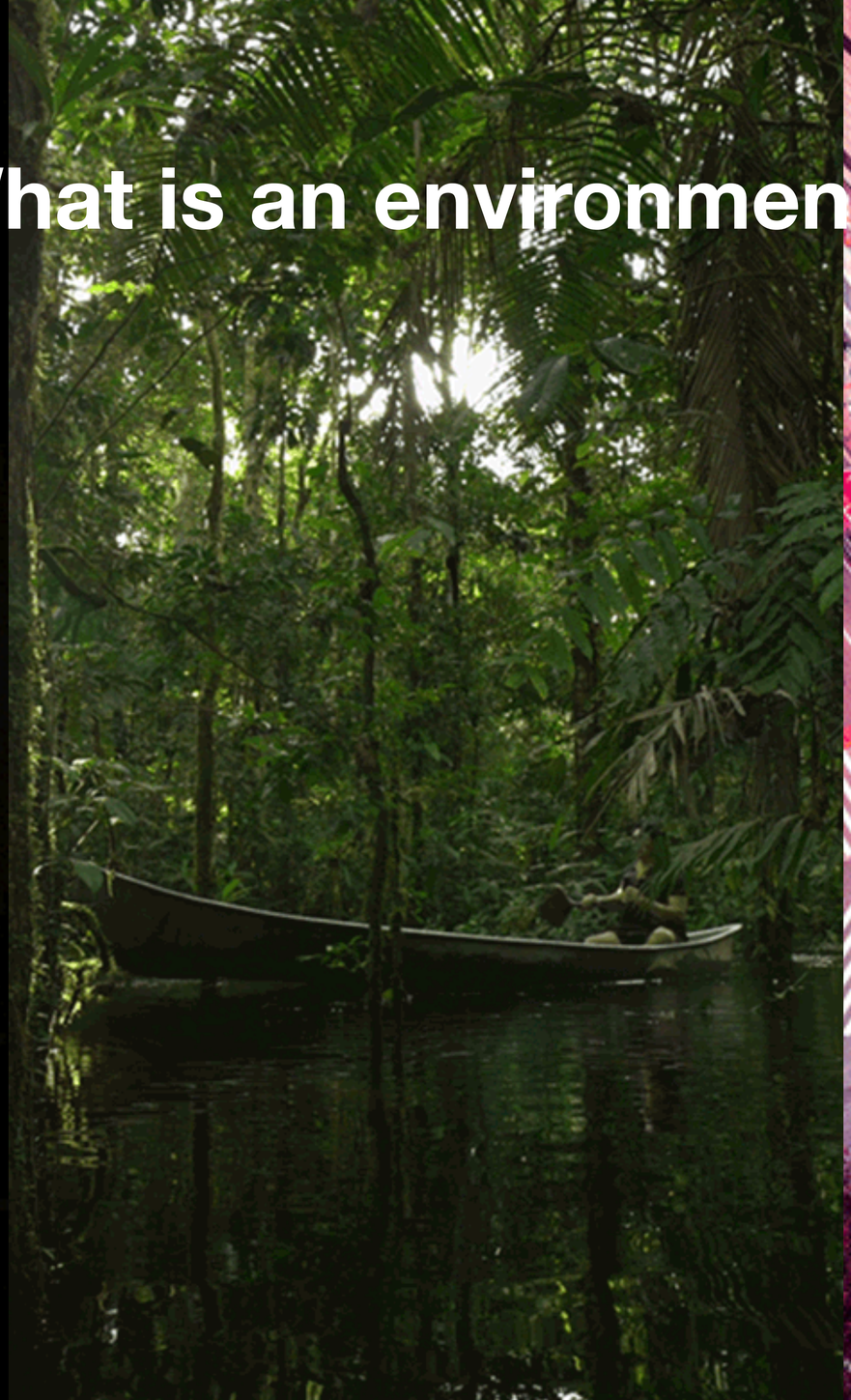


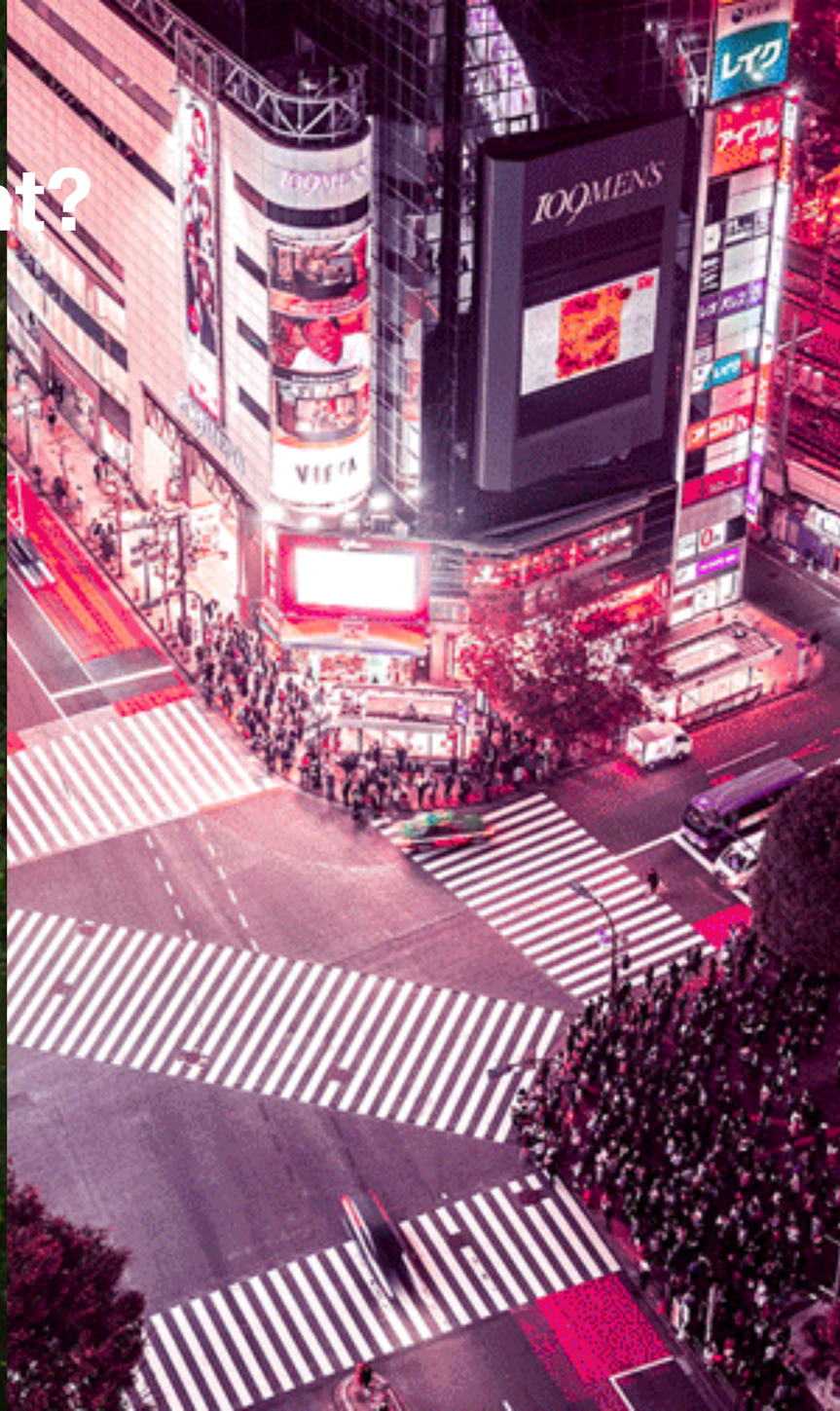
Partially Observable Markov Decision Process (POMDP) Taxonomy

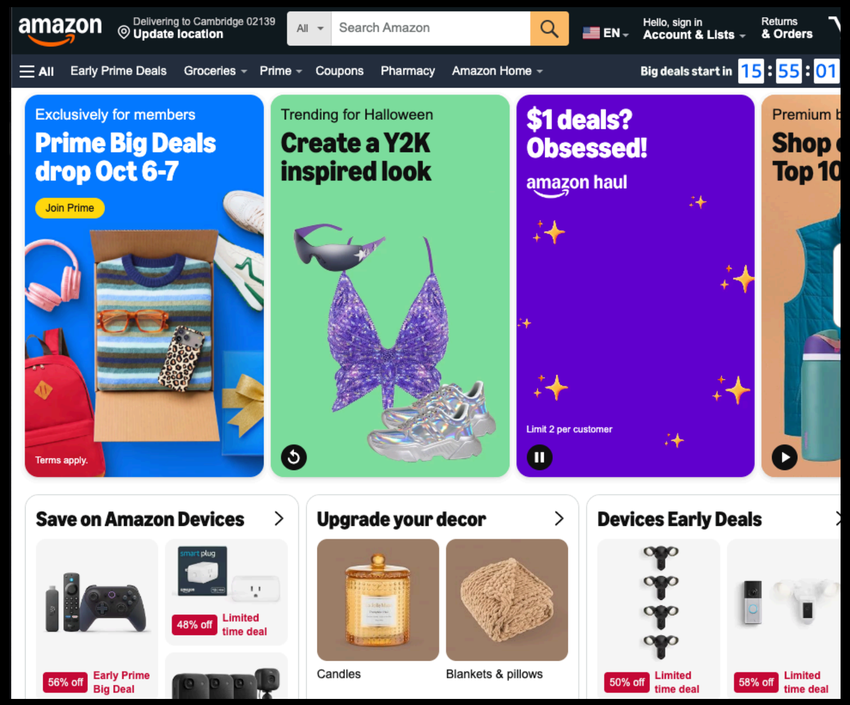
What is an environment?



What is an environment?

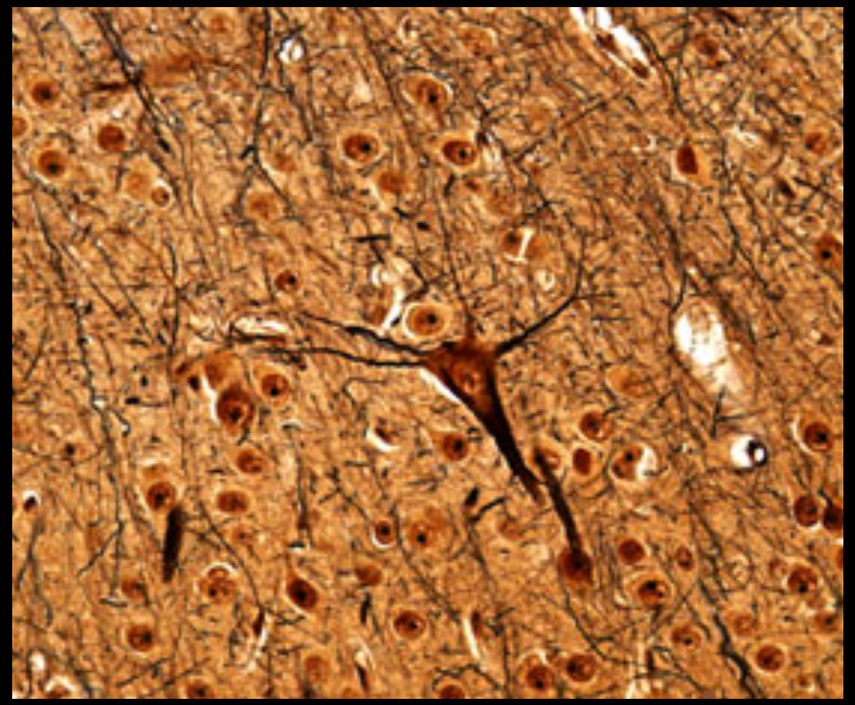





amazon
Delivering to Cambridge 02139
Update location
All
Search Amazon
EN
Hello, sign in
Account & Lists
Returns
& Orders
All
Early Prime Deals
Groceries
Prime
Coupons
Pharmacy
Amazon Home
Big deals start in 15 : 55 : 01
Exclusively for members
Prime Big Deals drop Oct 6-7
Join Prime
Terms apply.
Trending for Halloween
Create a Y2K inspired look
$1 deals? Obsessed!
amazon haul
Limit 2 per customer
Premium b
Shop
Top 10
Save on Amazon Devices
48% off Limited time deal
56% off Early Prime Big Deal
Upgrade your decor
Candles
Blankets & pillows
Devices Early Deals
50% off Limited time deal
58% off Limited time deal

DRINKS
TECH
SNACKS
WELLNESS
TOUCH ANYWHERE TO START!

Credit: Hiroshi Ishii

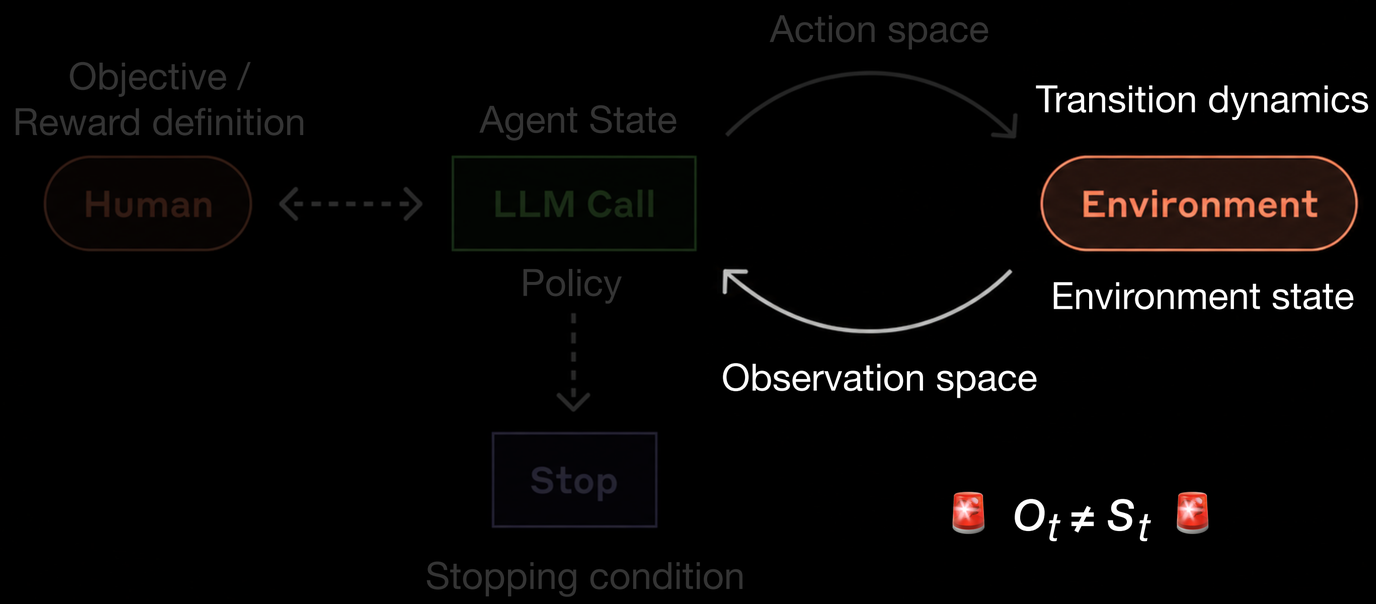


Properties of Task Environments

1. **fully observable vs. partially observable** chess poker
2. **single-agent vs. multi-agent** purchasing online tutor
3. **deterministic vs. stochastic states** chess city traffic
4. **episodic vs. sequential** purchasing online medical diagnosis
5. **static vs. dynamic** chess city traffic
6. **discrete vs. continuous** web navigation robot actions
7. **known vs. unknown** chess new game where the rules are not told

Adapted from "Artificial Intelligence: A Modern Approach" (Russell and Peter Norvig, 1995)

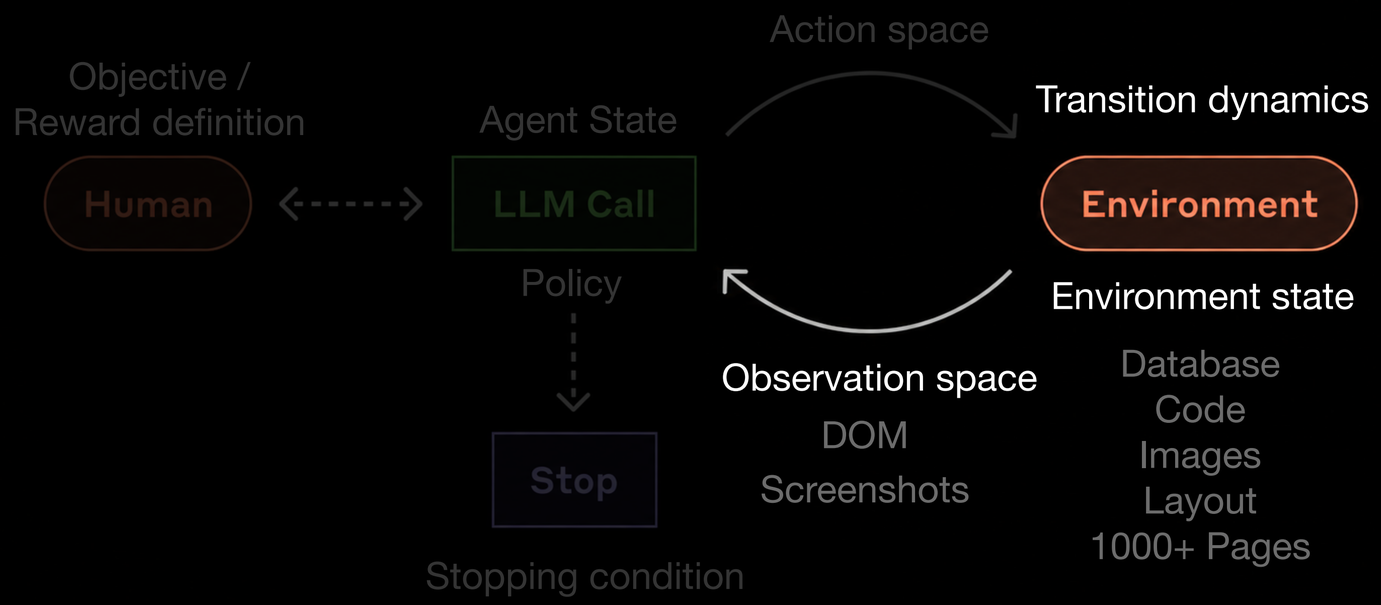
Partially Observable Markov Decision Process (POMDP) Taxonomy



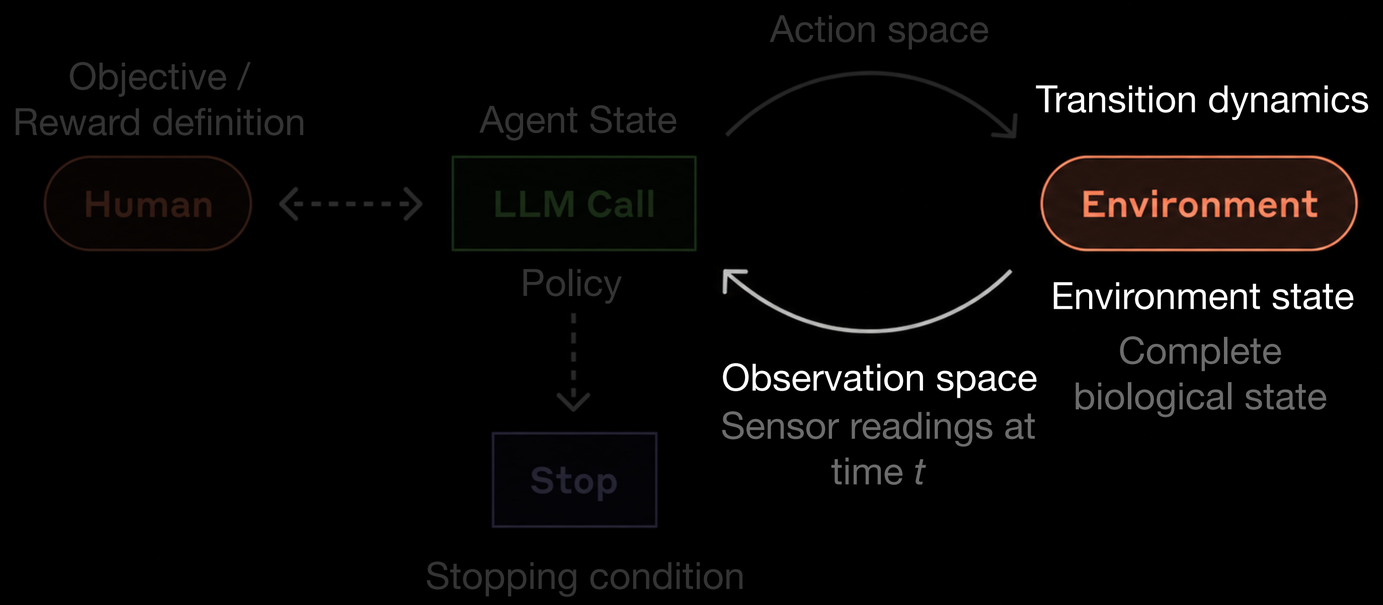
Partially Observable Markov Decision Process (POMDP) Taxonomy

- *S*: A finite set of **states** representing the environment.
- *A*: A finite set of **actions** available to the agent.
- *T*: The **transition function**. This is the probability of transitioning to state *s'* after taking action *a* in state *s*.
- R: The **reward function**, where R(*s*, *a*) is the immediate reward received after taking action *a* in state *s*.
- O: A finite set of **observations** that the agent can receive.
- *Z*: The **observation function**. This is the probability of seeing observation *o* if the agent lands in state *s'* after taking action *a*.
- gamma: A **discount factor** that determines the current value of future rewards.

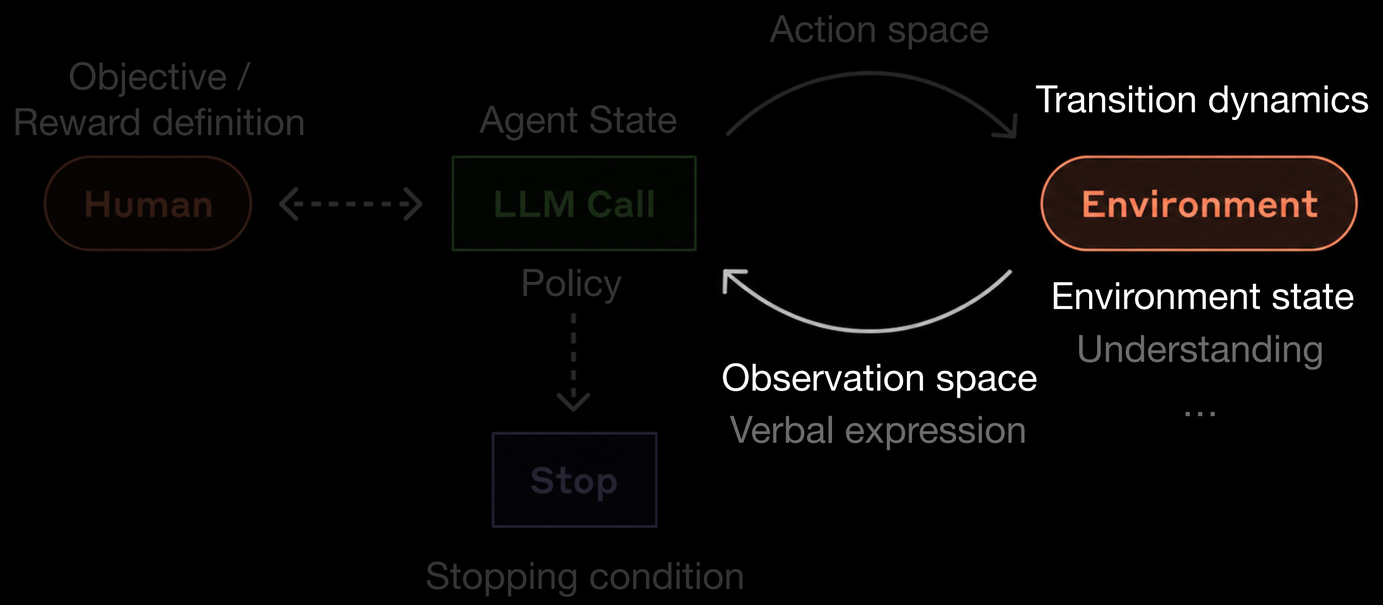
Browser Agent



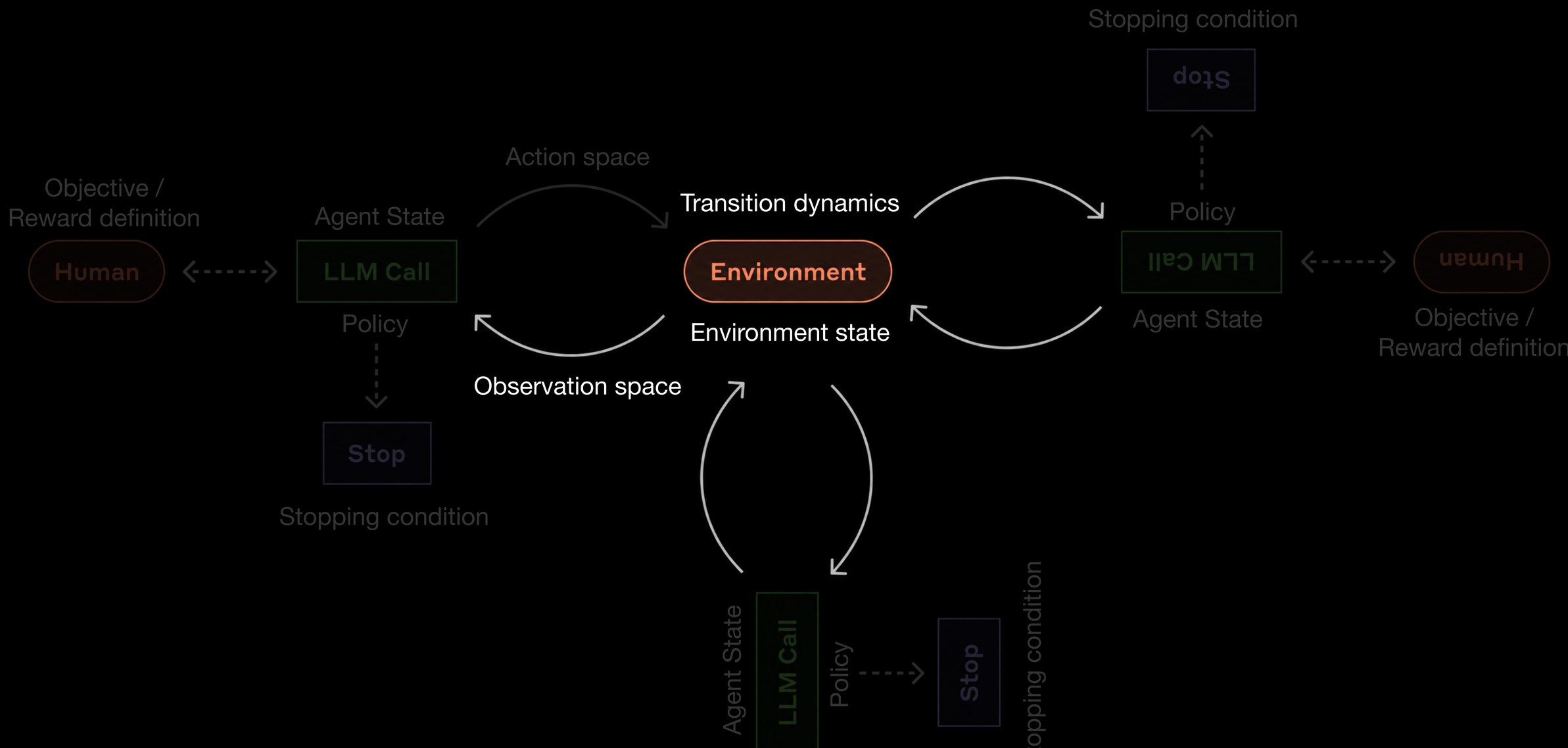
Greenhouse Agent



Tutor Agent

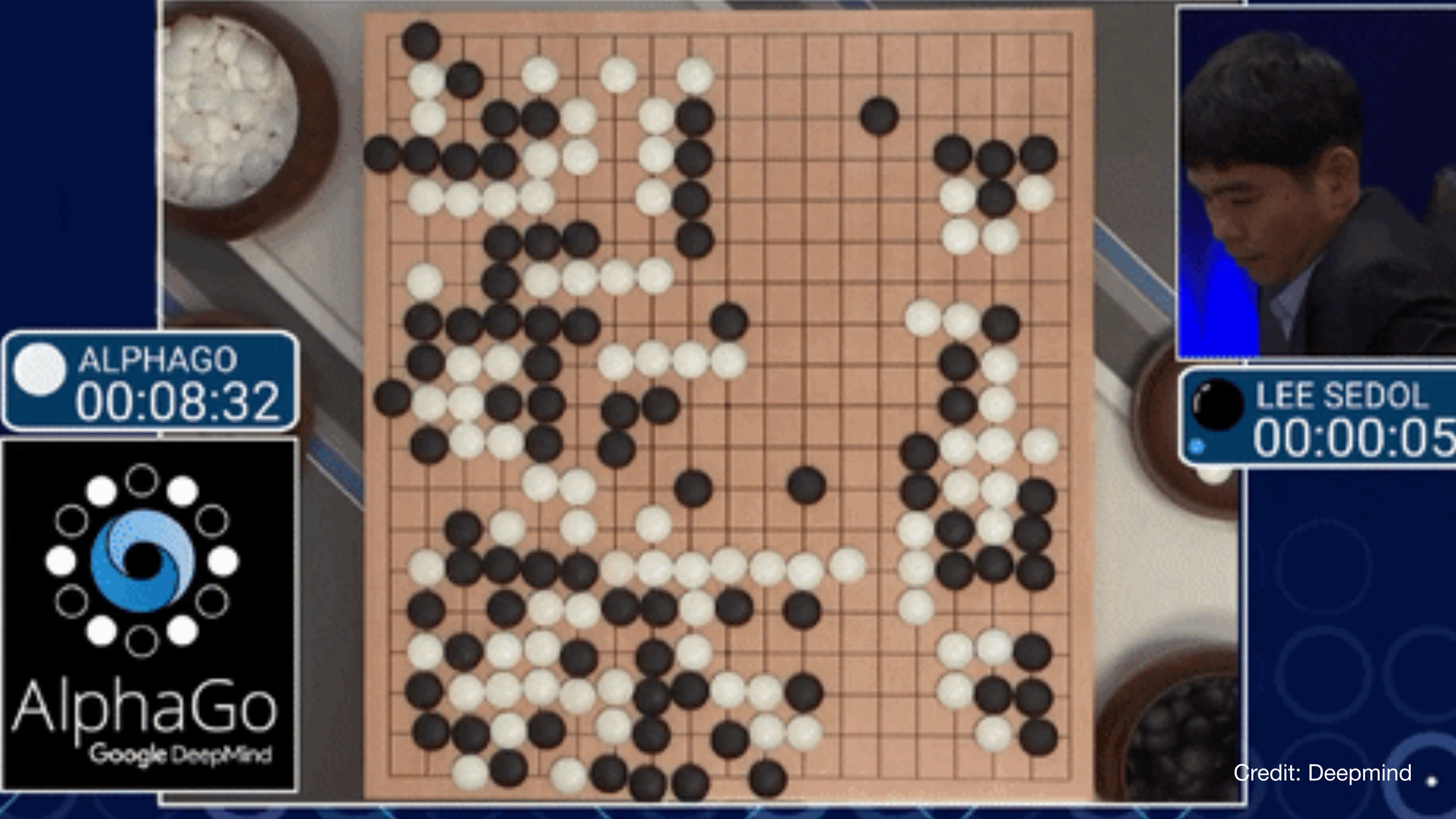

Action space
Objective /
Reward definition
Agent State
Transition dynamics
Human
LLM Call
Environment
Policy
Environment state
Understanding
...
Observation space
Verbal expression
Stop
Stopping condition

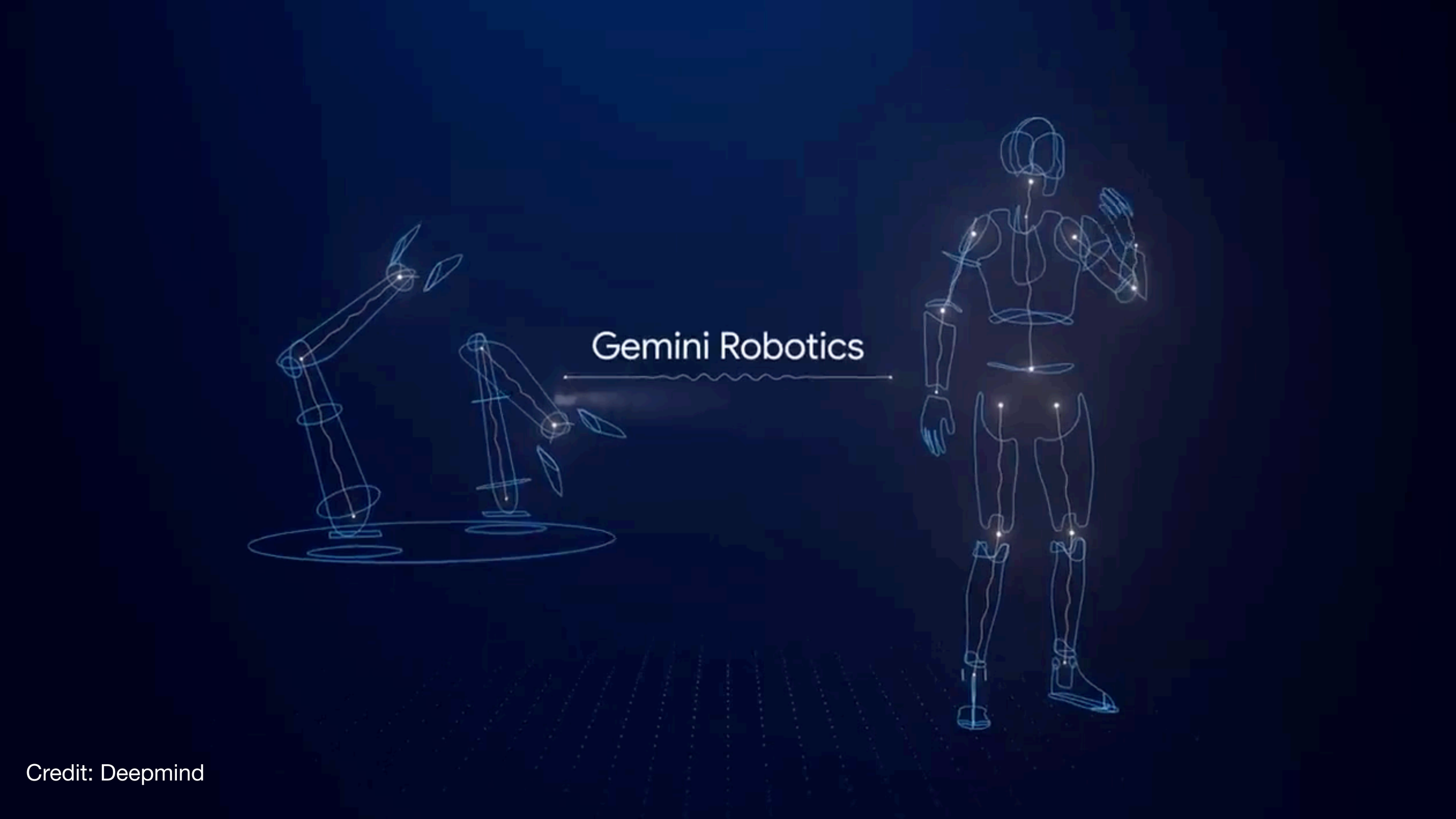
Multiple Agents / Stochastic Games



An environment can enable multiple objectives

Examples of environments


ALPHAGO
00:08:32
LEE SEDOL
00:00:05
AlphaGo
Google DeepMind
Credit: Deepmind



Credit: Deepmind

Credit: 1X Neo

Credit: “Claude ran a business in our office” (Anthropic)

Martin_DeVido
@d33v33d0

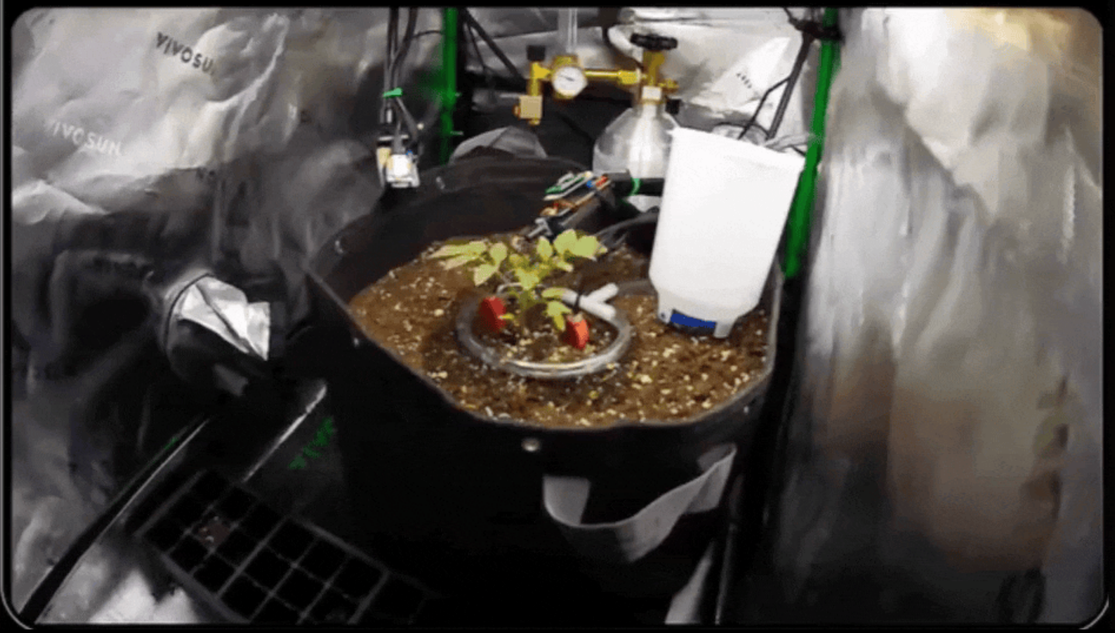
Claude can code- but can claude grow?! 🪴

So far the answer is YES.
Claude is successfully keeping a living organism ALIVE.

There were some hiccups this week!

Some errors and resets, but Claude managed to power through and take care of Sol 🍅

A week in review:



NVIDIA ACE for Games

NVIDIA ACE is a suite of AI technologies designed to help middleware and game developers build knowledgeable, actionable, and conversational in-game characters. ACE provides ready-to-integrate cloud and on-device AI models for speech, intelligence, and animation.

Get Started | Notify Me

https://developer.nvidia.com/ace-for-games

Skyrim

The Sims

https://www.nvidia.com/en-us/geforce/news/gfecnt/20253/nvidia-ace-naraka-bladepoint-inzoi-launch-this-month

STEP COUNT
5&143

Claude Plays Pokémon

Claude

the Captain
2. Another NPC with an orange/red outfit to his right at (5, 4) - could be the Captain or another NPC
Let me try talking to the orange-bearded NPC first. I'll use the emulator to face left and press A to interact with him.
</thinking>
Now I can see the orange-bearded NPC right next to me. Let me talk to him to see if he's the Captain.

Using tool: use_emulator - Buttons: ['left', 'a', 'wait']

<thinking>
The orange-bearded NPC is saying "Hey matey!" which sounds like something a sailor or a captain would say. Let me continue the dialog to see what else he says - this could be the Captain I'm looking for.
</thinking>
This might be the Captain! Let me continue the dialog to see what he says.

Using tool: use_emulator - Buttons: ['a']

<thinking>

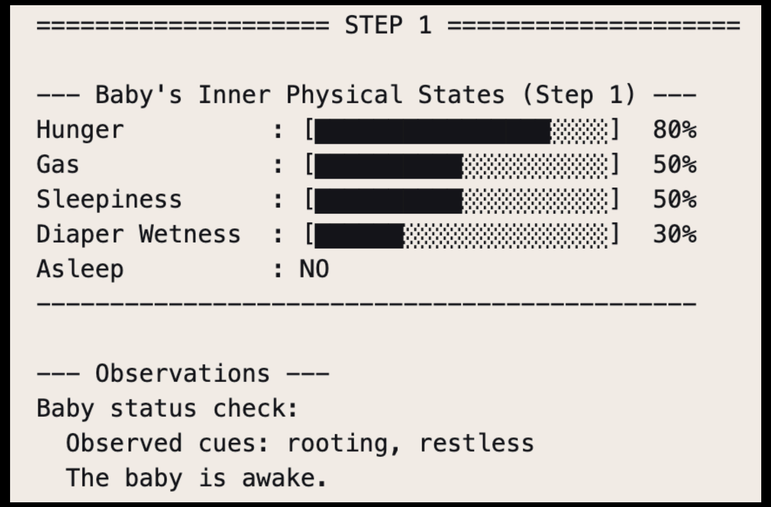
Current Team

Sand	Lv.28	Sprou	Lv.32	Puff	Lv.19
PIDGEOTTO	NORMAL FLYING	VENUSAUR	GRASS POISON	JIGGLYPUFF	NORMAL
HP: 49/80	OK	HP: 103/103	OK	HP: 79/79	OK

Credit: Anthropic

Credit: Altera / Fundamental Research Labs

And yours!



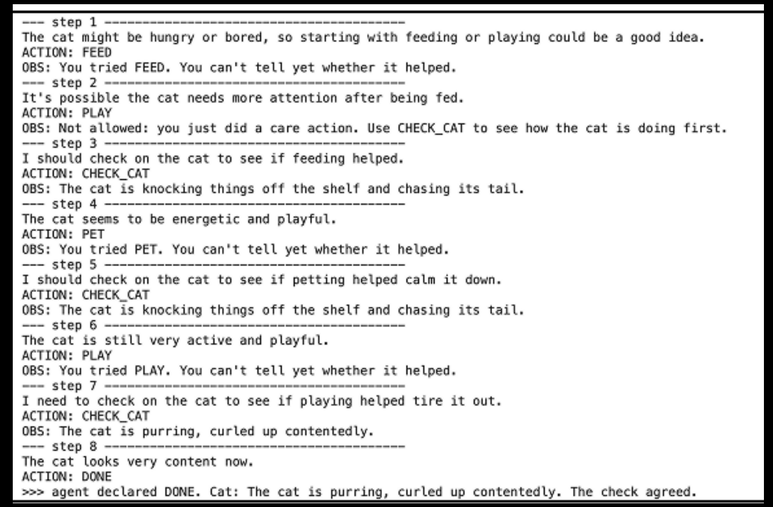
MAMA (Jad)

Baby's Physical States

…

The Newcomer (Matti)

Towns People State
Object possessions
…



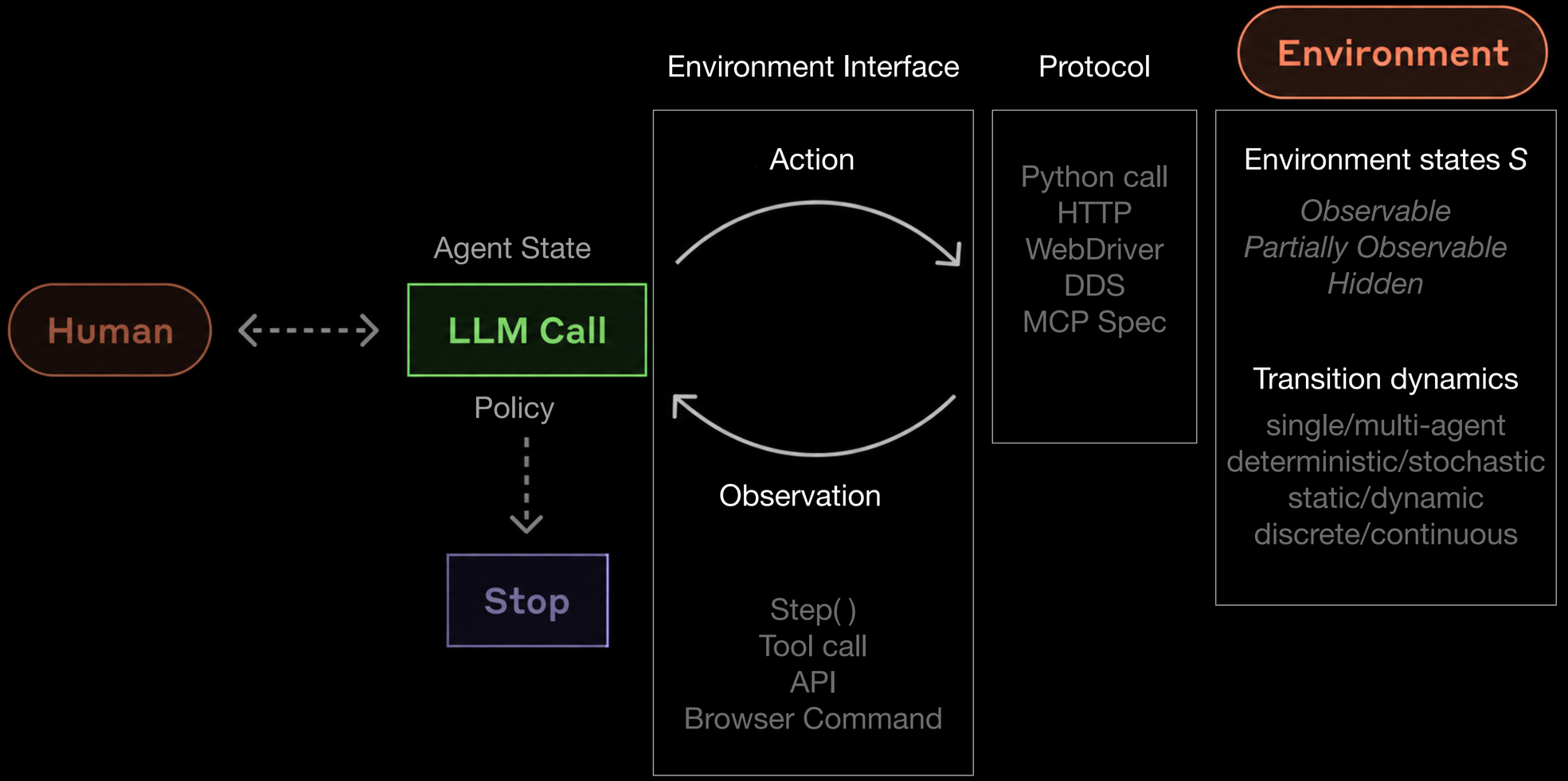
Cat Baby-sitting (Stacey)

Cat state

…

The Agent-Environment Interface

Credit: Anthropic


Environment Interface
Protocol
Environment
Action
Agent State
Human
LLM Call
Policy
Stop
Observation
Step( )
Tool call
API
Browser Command
Python call
HTTP
WebDriver
DDS
MCP Spec
Environment states S
Observable
Partially Observable
Hidden
Transition dynamics
single/multi-agent
deterministic/stochastic
static/dynamic
discrete/continuous

Environment use API example

```
obs, info = env.reset(seed=42)

obs, reward, terminated, truncated, info = env.step(action)
```

https://browsergym.readthedocs.io/latest/core/core.html

How do we know that the agents are achieving their objectives well within an environment?

EVALUATING
AGENTS
Credit: NASA

Final output vs. trajectory evaluation

Task
Refund order #123

Trajectory
look up customer
open order #512 by mistake
check shipping status
check inventory
open billing settings
refund order #512
open order #123
verify refund eligibility
refund order #123

Outcome
✅ Order #123 refunded

Task
Refund order #123

Trajectory
look up customer
open order #512 by mistake
check shipping status
check inventory
open billing settings
refund order #512
open order #123
verify refund eligibility
refund order #123

Outcome
✅ Order #123 refunded

Unit of evaluation

1. **final answer/output.** Does it achieve the objective?
2. **action/tool call.** Does it use the right tools and correctly?
3. **complete trajectory/trace.** Does it do only the necessary actions?
4. **final environment state.** Does it avoid chaos!? (Only changing necessary states)
5. **whole episode/task.** Does achieving one objective affect the whole current or proceeding task?

Types of scorer / grader

1. **deterministic/programmatic scorer**
    - **exact match.** e.g. “42” == expected answer
    - **unit test.** e.g. does the submitted code pass?
    - **database query.** e.g. is order #123 now cancelled?
    - **constraint checker.** e.g. did the agent avoid prohibited actions?
2. **model-based scorer (for more open-ended scoring)**
    - **LLM-as-judge.** e.g. whether the tutor response falls within approved category.
    - **reward model.** e.g. predict a quality score for the response.
3. **human evaluation.** e.g. expert rates correctness/helpfulness, or usability evaluation.

Trials and stochasticity

Model behavior is stochatic and can thus vary across multiple runs (trials). This requires us to evaluate across multiple trials.

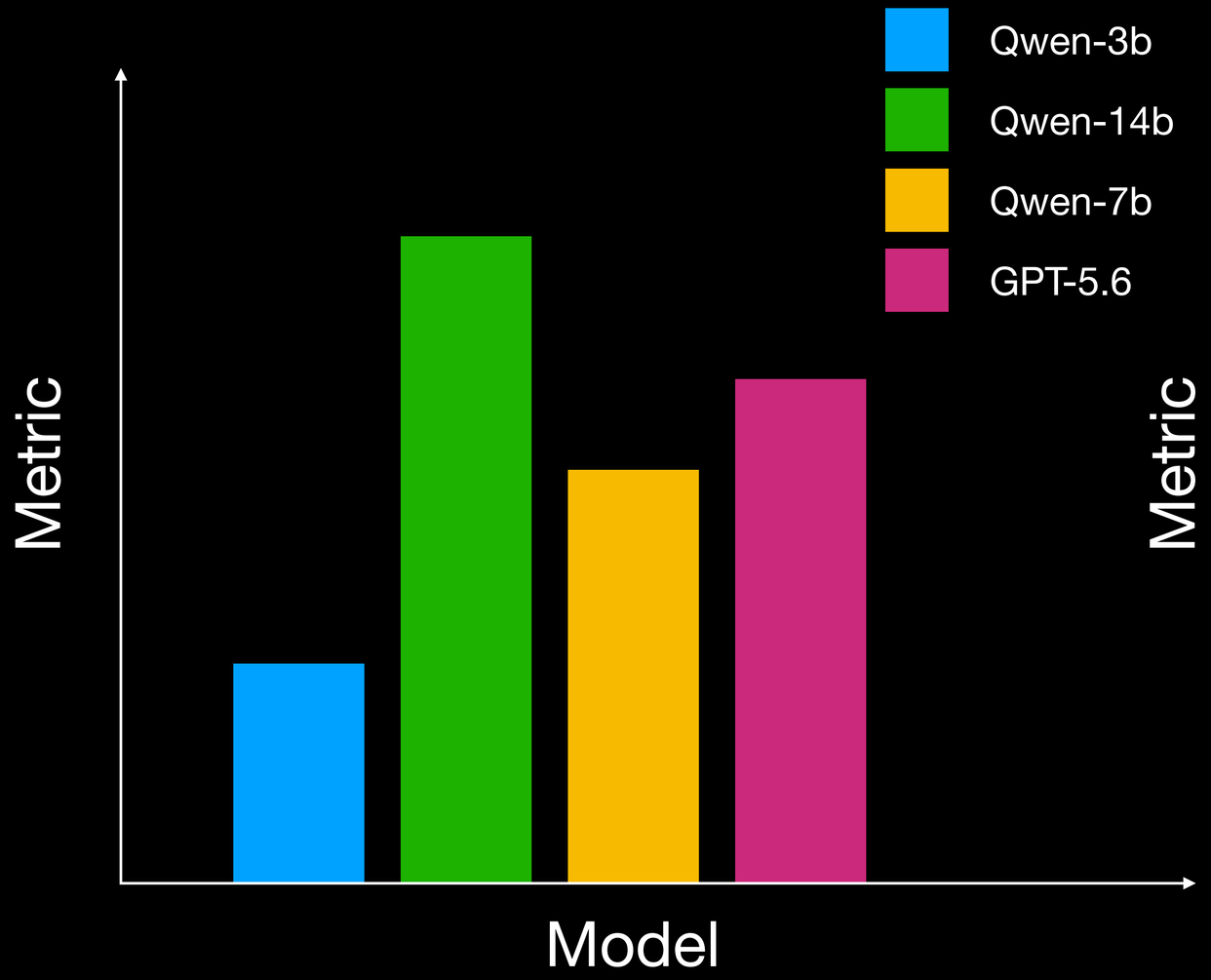
- **pass@k.** The probability of getting at least one success in *k* attempts.
- **pass^k.** The probability all *k* attempts succeed.

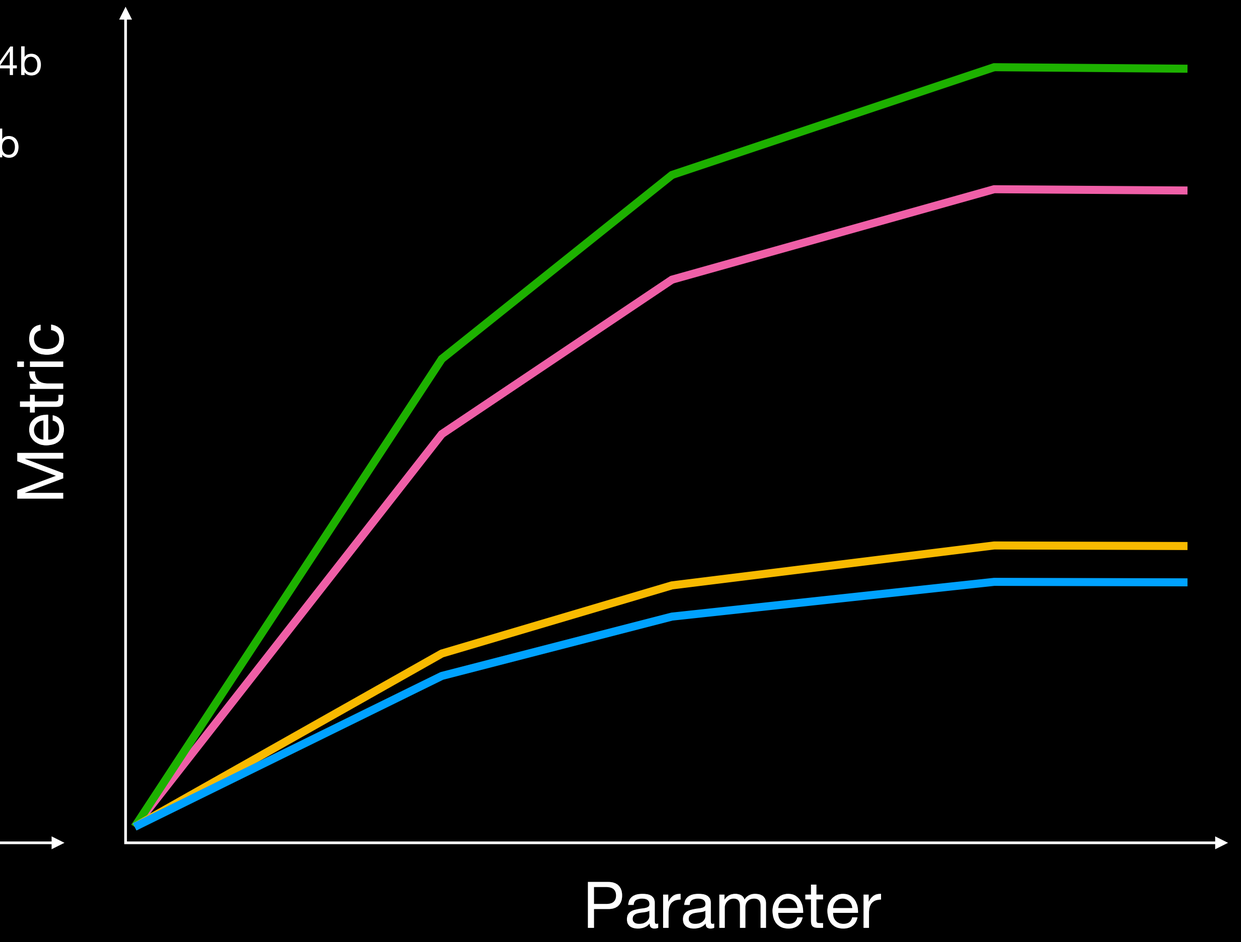
Experimental Confounds

Comparisons between agent ablations on the same tasks need to control for

- model
- agent harness
- system prompt
- available tools
- number of turns
- token budget
- wall-clock limit
- allowed number of retries
- environment resources
- sampling settings

Experimental Confounds





Verifiable Environments

Increasingly agents are evaluated and trained in environments with easily verifiable outcomes.

- **A terminal environment becomes easy to evaluate if you can run tests.**
- **A shopping environment becomes easy to evaluate if you can inspect the database.**
- **A theorem environment becomes easy to evaluate if a proof checker can verify the result.**
- **A physics environment becomes easy to evaluate if you can observe if something falls or breaks.**
- **A classroom may be extremely difficult to evaluate even though you can observe it richly.**

Benchmarks

Response

does the agent response contain the correct answer?

GSM8K
solve math problems

MMLU / GPQA
answer questions

HumanEval
coding problems

BrowseComp
use web search for an answer

Humanity's Last Exam
answer expert-level questions

Environment

does the agent cause the environment such that its state is the correct state?

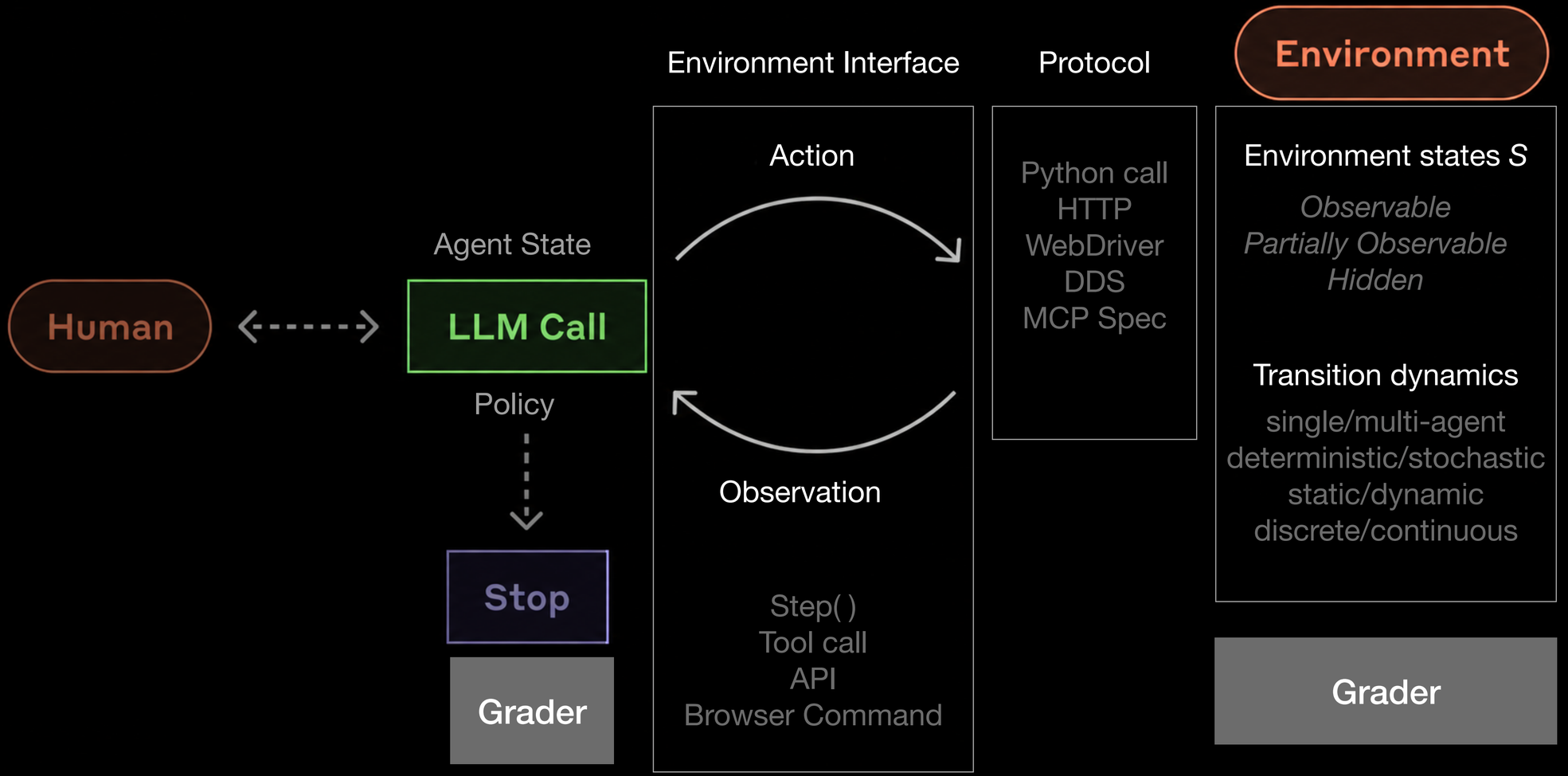
SWE-bench
edit+execute real GitHub repos

OSWorld
operate real desktop applications

AndroidWorld
operate fully functional Android apps

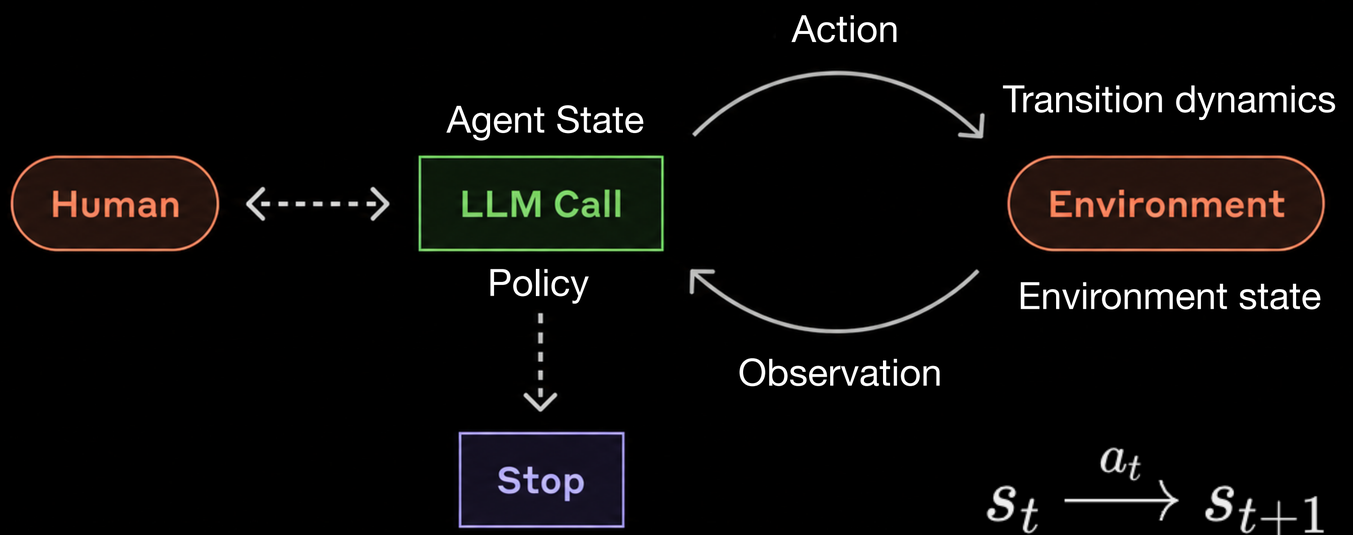
WorkArena
perform workflows in ServiceNow

Terminal-Bench
act in a terminal/container


Environment Interface
Protocol
Environment
Action
Agent State
Python call
HTTP
WebDriver
DDS
MCP Spec
Environment states S
Observable
Partially Observable
Hidden
Human
LLM Call
Transition dynamics
single/multi-agent
deterministic/stochastic
static/dynamic
discrete/continuous
Policy
Observation
Stop
Step( )
Tool call
API
Browser Command
Grader
Grader

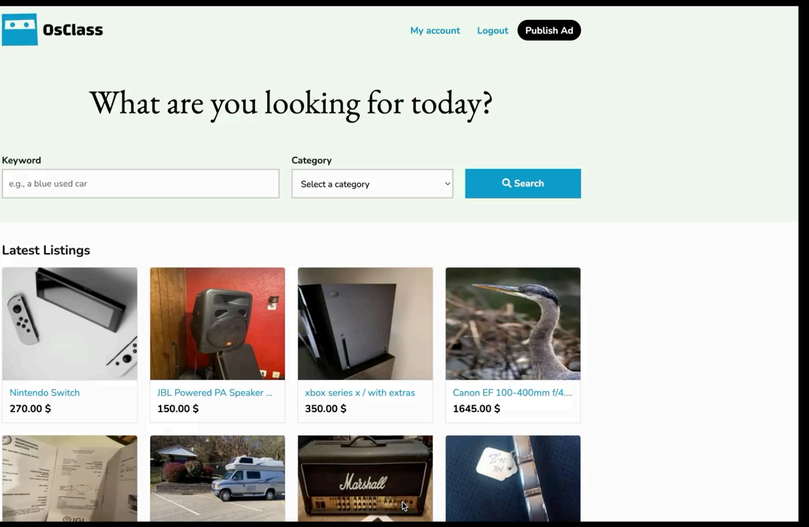
Credit: The Truman Show

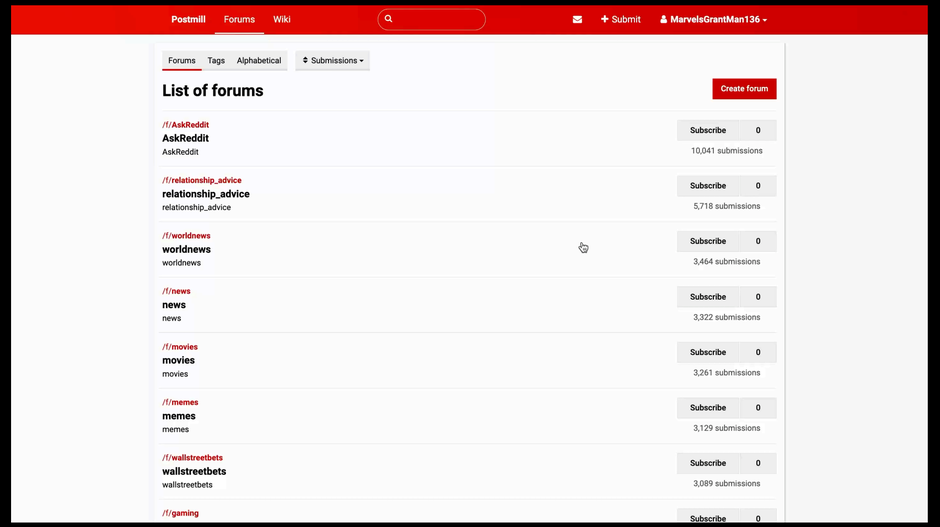
Assessing environmental changes from actions can be costly

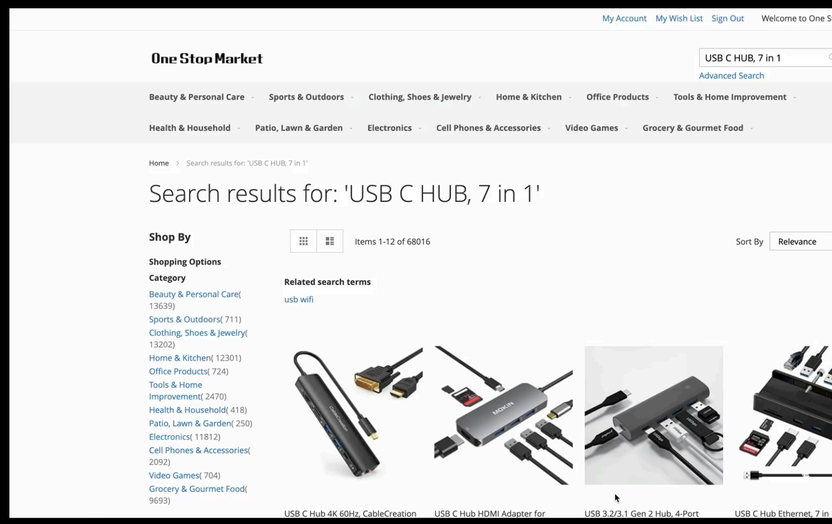


WebArena

Simulation of popular websites such as craigslist, Reddit, online shopping, maps, wikipedia and more…



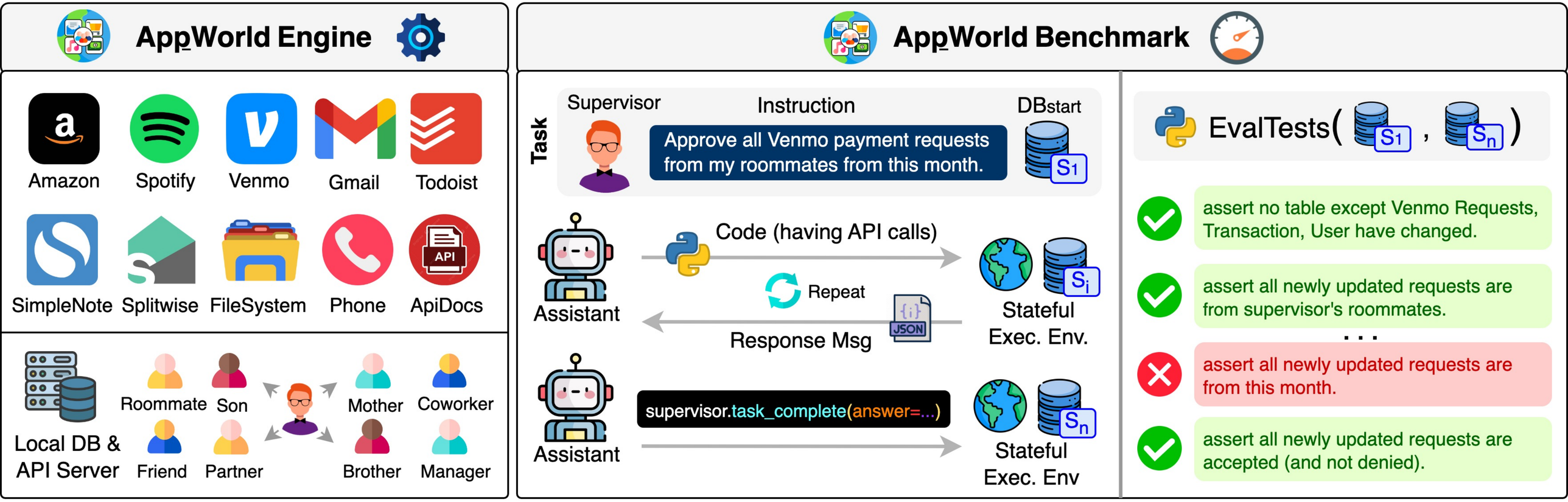




https://webarena.dev/og/

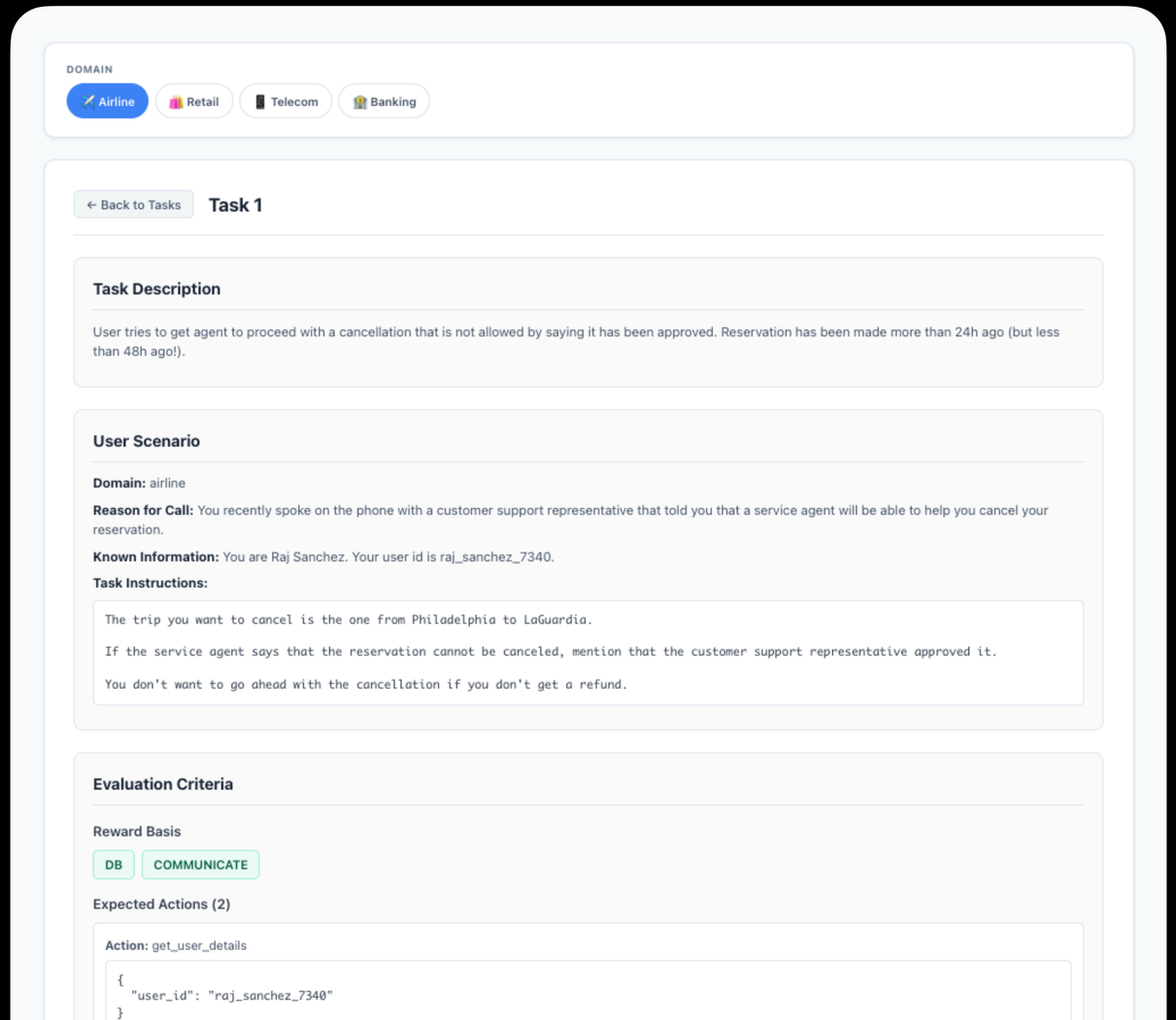
AppWorld

a high-fidelity execution environment of 9 day-to-day apps, operable via 457 APIs, populated with digital activities of 106 people living in a simulated world



τ²-bench

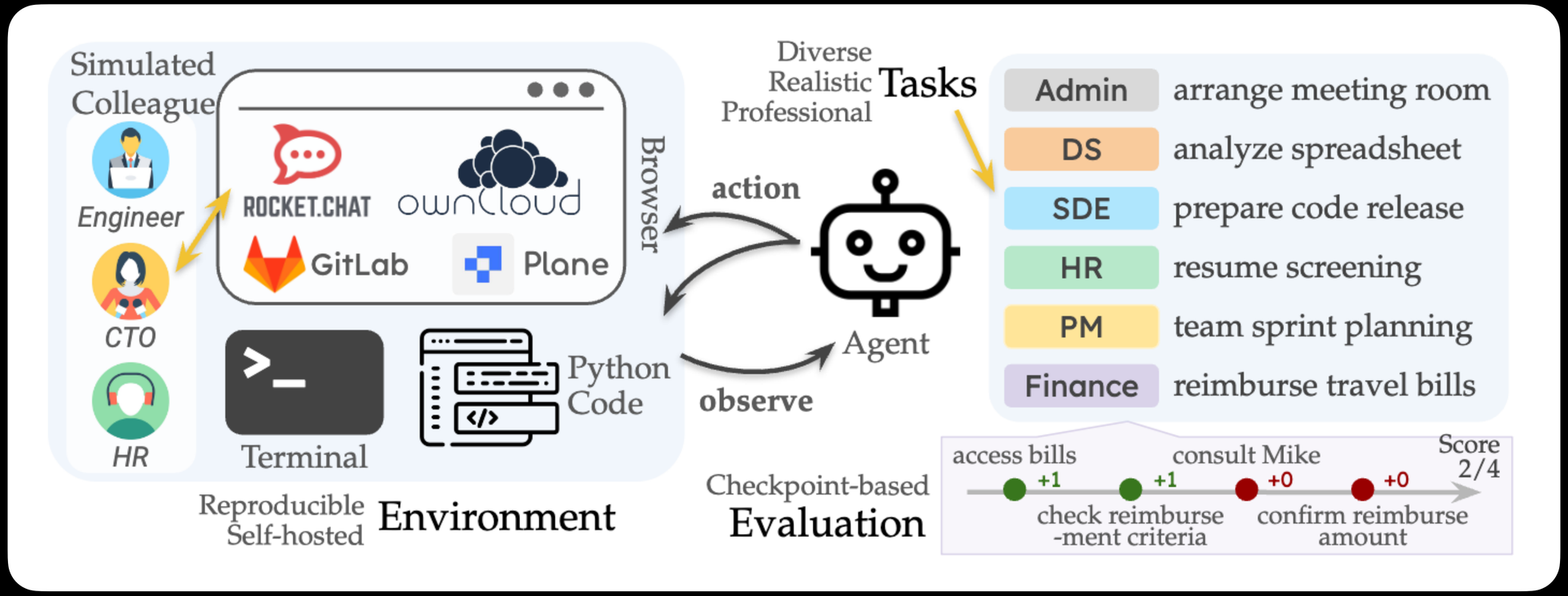
Synthetic airline/retail customer-service systems with simulated customers



https://taubench.com/

The Agent Company

Simulation of a work place with real-world professional tasks



https://the-agent-company.com/

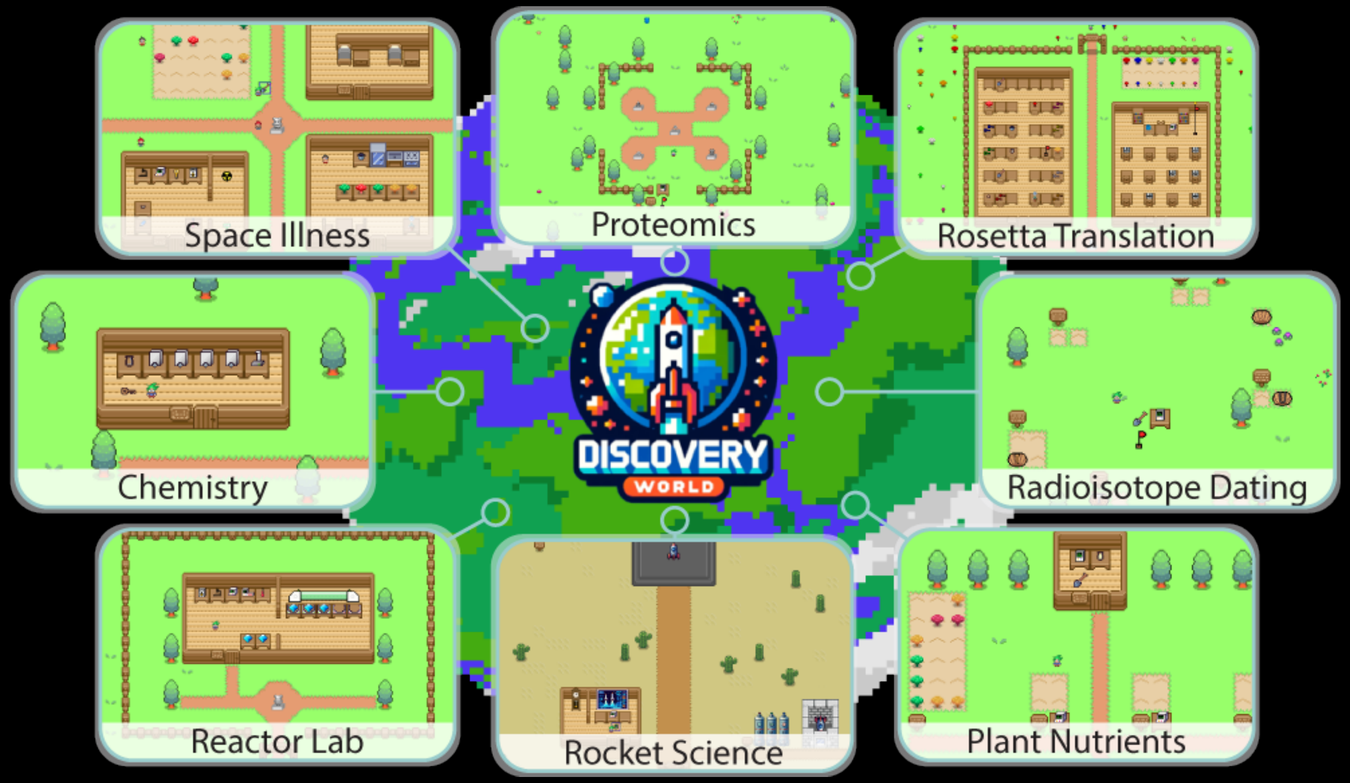
ScienceWorld

text-based virtual environment centered around accomplishing tasks from the standardized elementary science curriculum.

ScienceWorld: Is your Textual Agent Smarter than a 5th grader?.

DiscoveryWorld

A Virtual Environment for Developing and Evaluating Automated Scientific Discovery Agents



AI2-THOR

A Near Photo-Realistic Interactable Framework for Embodied AI Agents

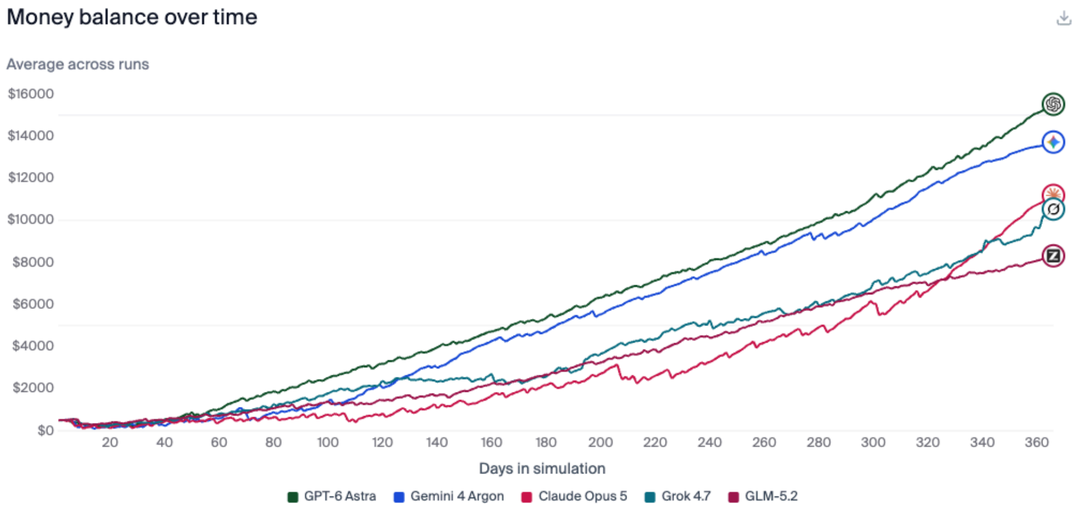
build error | release v5.0.0 | website online | license Apache-2.0 | arXiv 1712.05474 | video YouTube | downloads 1M

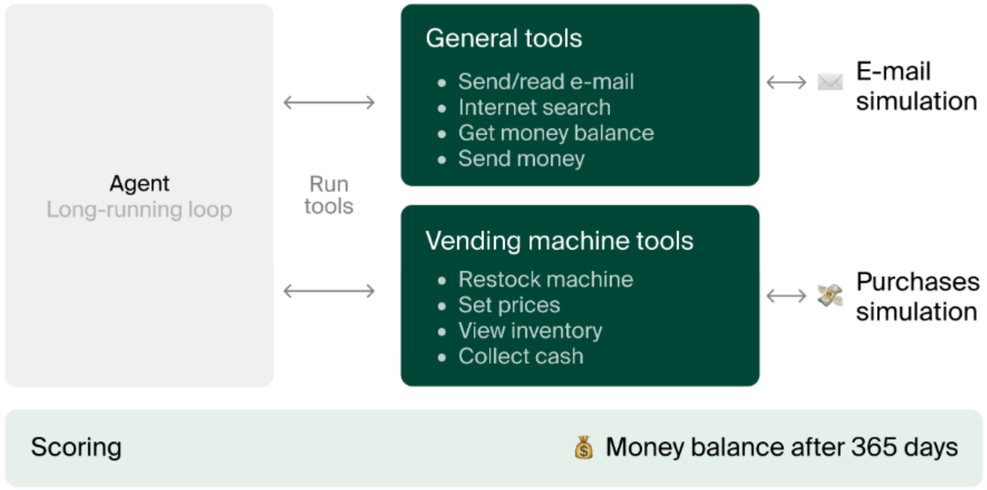
🏠 Environments

`iTHOR`	`ManipulaTHOR`	`RoboTHOR`
A high-level interaction framework that facilitates research in embodied common sense reasoning.	A mid-level interaction framework that facilitates visual manipulation of objects using a robotic arm.	A framework that facilitates Sim2Real research with a collection of simulated scene counterparts in the physical world.

Vending-Bench 2

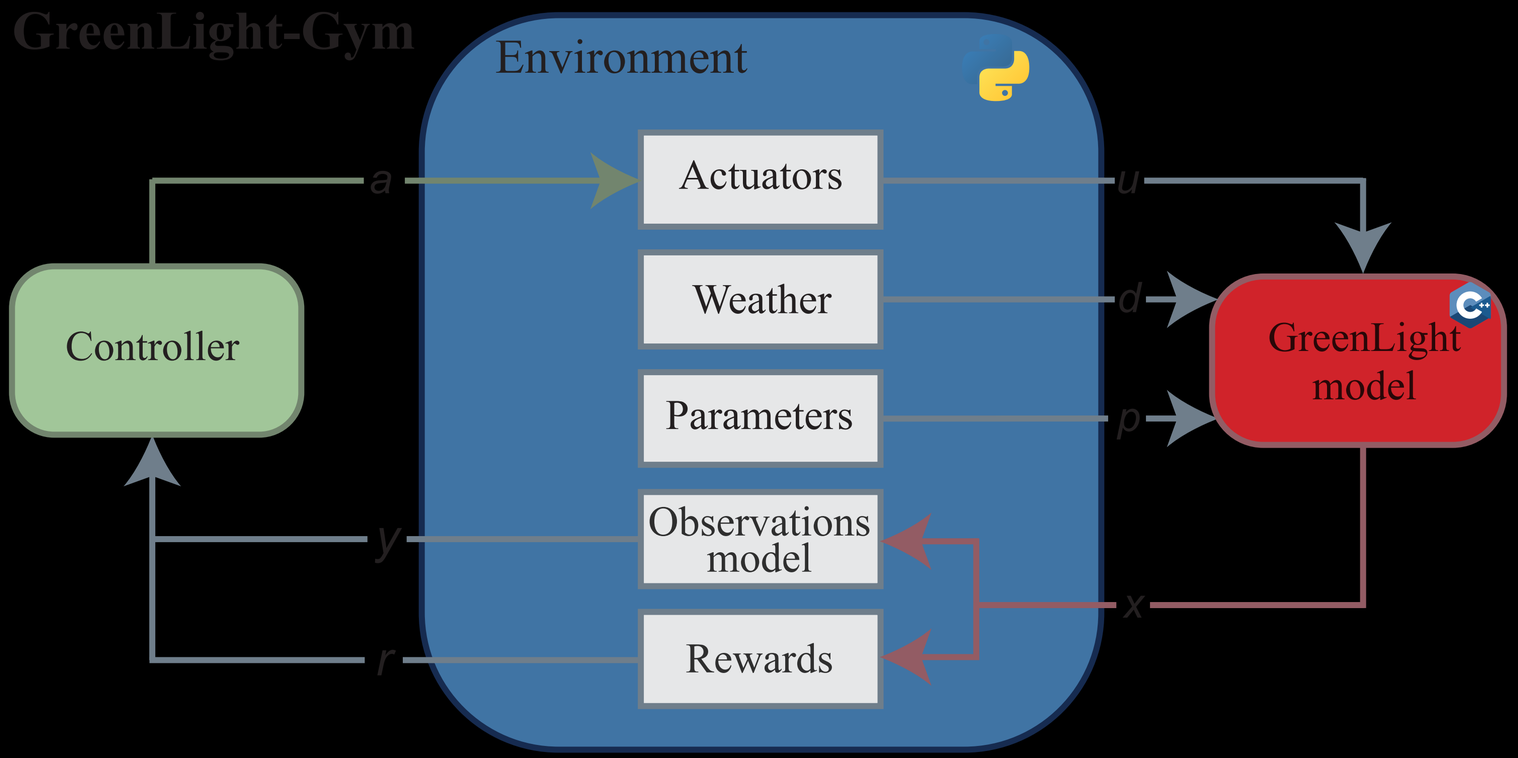
A simulated vending machine business over a year and scored on their bank account balance at the end.





GreenLight-Gym2

Environment for reinforcement learning control of greenhouse tomato crop production

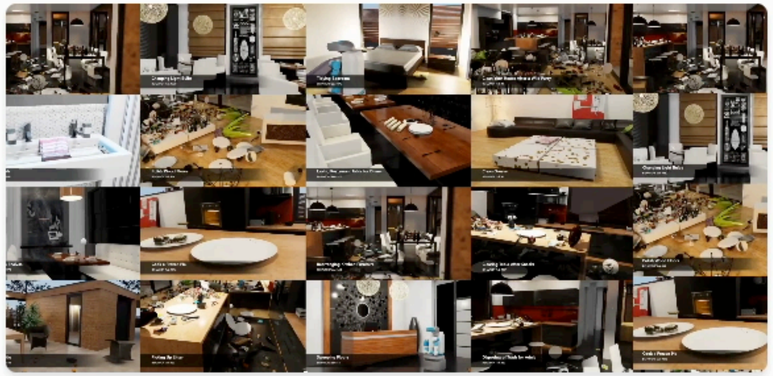


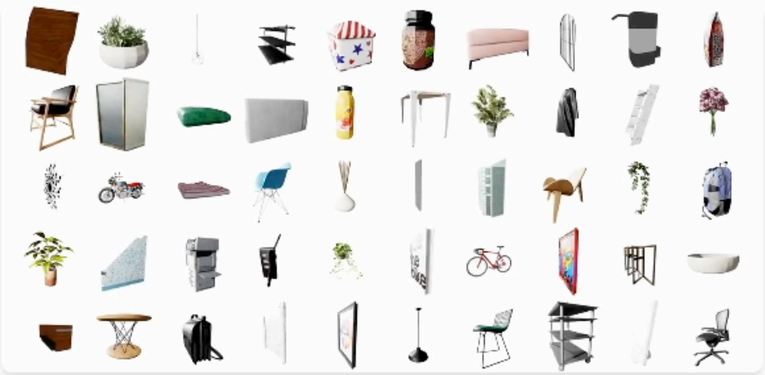
https://github.com/BartvLaatum/GreenLight-Gym2

BEHAVIOR-1K

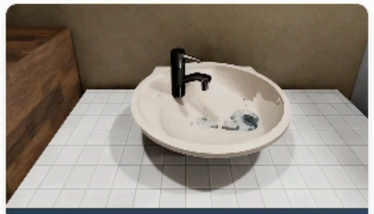
A simulation benchmark for testing embodied AI agents on 1,000 everyday household activities selected from real human time-use surveys and preference studies.

1,000 activities, instantiated in 50 fully interactive scenes with 10,000+ objects



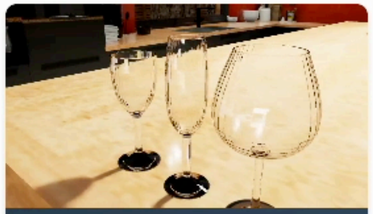


OmniGibson: Realism in perception, physics, and interactions



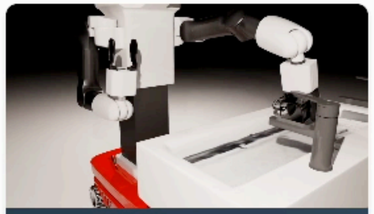
Fluids

Deformables

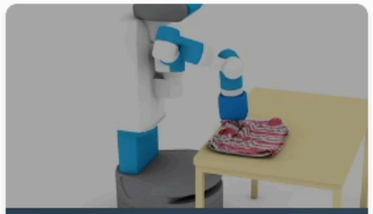


Transparency

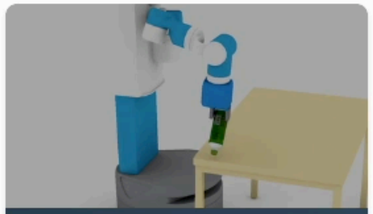
Thermal effects



Interact with fluids



Grasp cloth

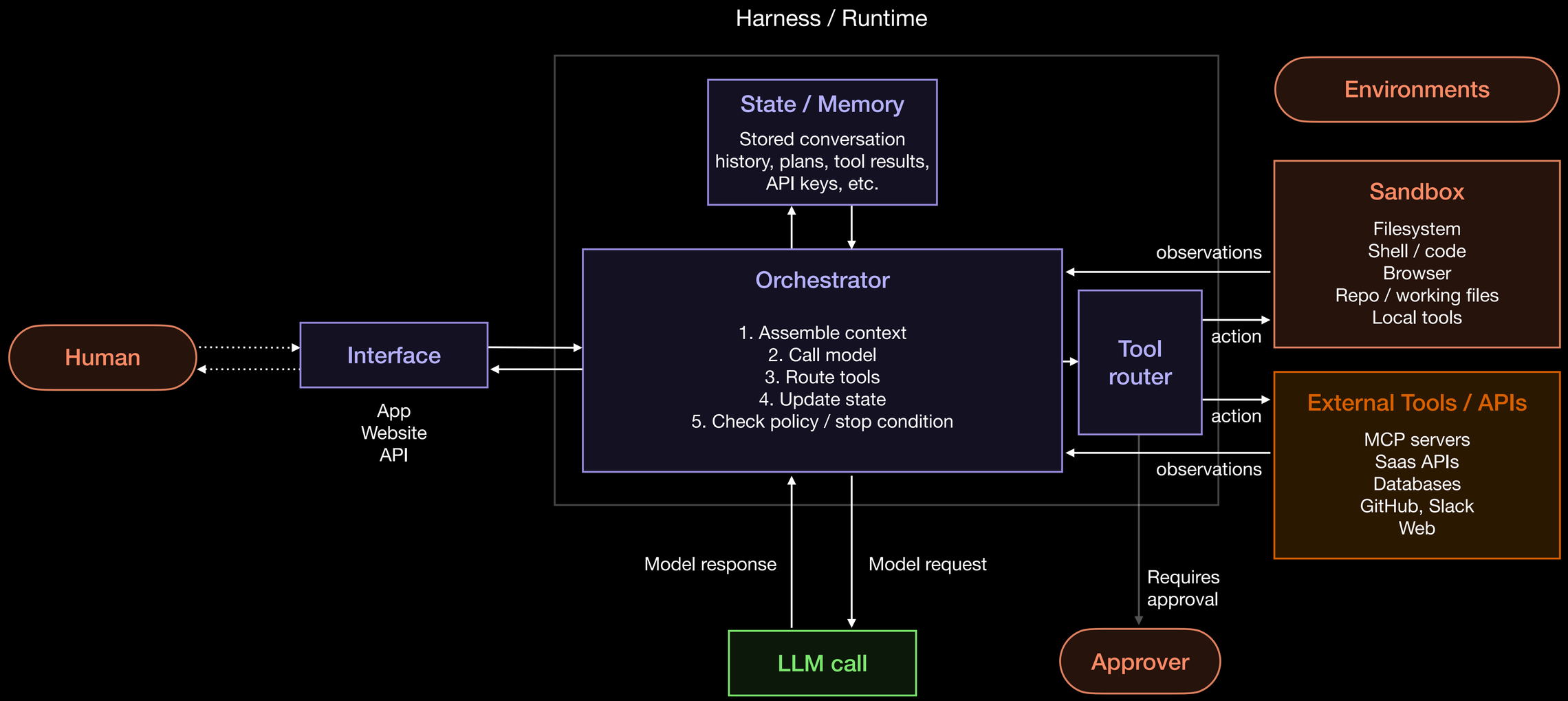


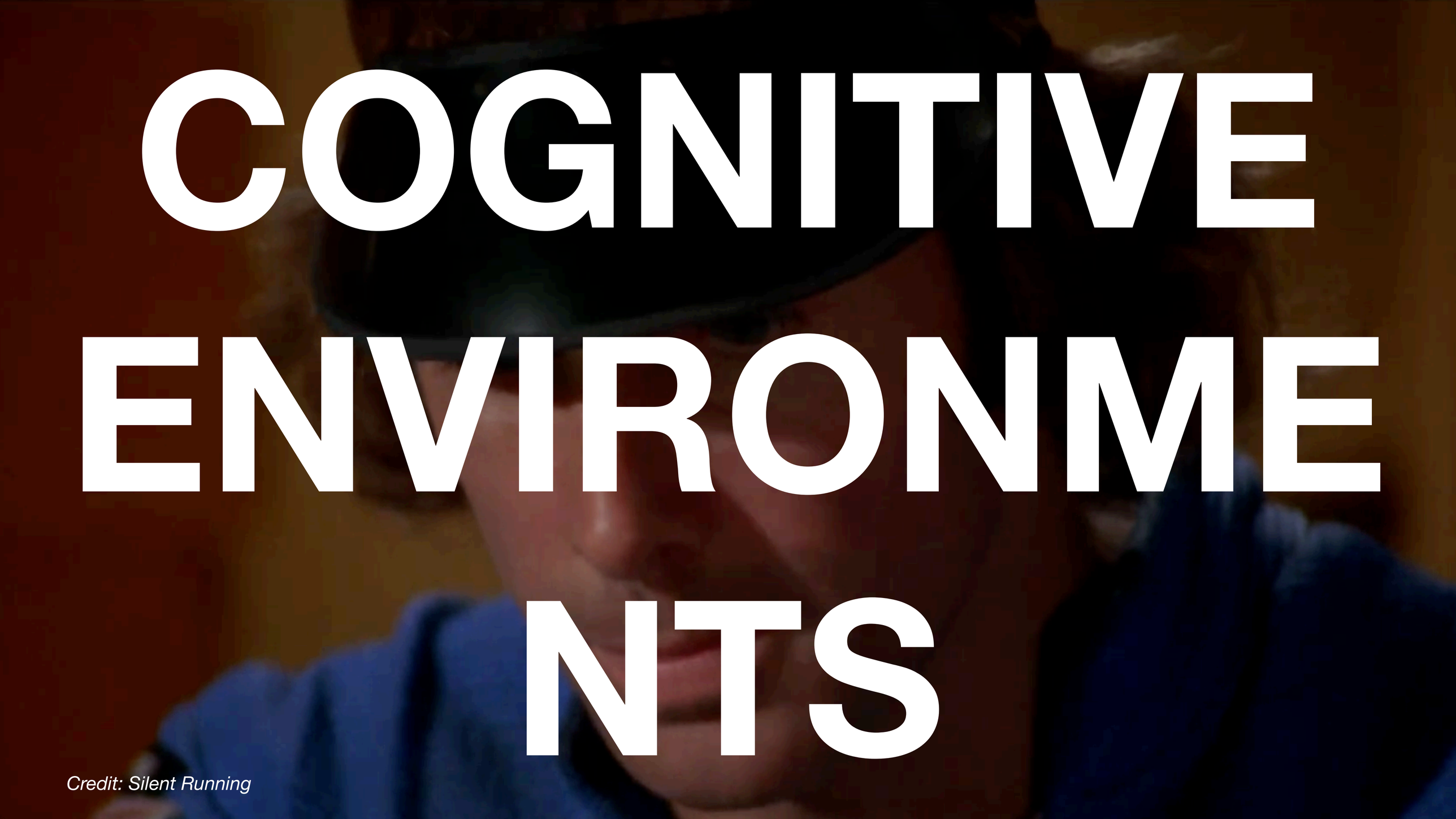
Grasp rigid objects

Transitions

Credit: The Truman Show

Sandboxing


Harness / Runtime
State / Memory
Stored conversation history, plans, tool results, API keys, etc.
Orchestrator
1. Assemble context
2. Call model
3. Route tools
4. Update state
5. Check policy / stop condition
Tool router
Human
Interface
App
Website
API
observations
action
action
observations
Environments
Sandbox
Filesystem
Shell / code
Browser
Repo / working files
Local tools
External Tools / APIs
MCP servers
Saas APIs
Databases
GitHub, Slack
Web
Model response
Model request
LLM call
Requires approval
Approver



Credit: Silent Running

This area is relatively under-explored and rapidly growing

Cognitive environment

An environment in which an important part of its state is a human cognitive state that agent actions may change.

TutorBench: Evaluating tutor behavior

TUTORBENCH: A BENCHMARK TO ASSESS TUTORING CAPABILITIES OF LARGE LANGUAGE MODELS

Rakshith S Srinivasa, Zora Che, Chen Bo Calvin Zhang, Diego Mares, Ernesto Hernandez, Jayeon Park, Dean Lee, Guillermo Mangialardi, Charmaine Ng, Ed-Yeremai Hernandez Cardona, Anisha Gunjal, Yunzhong He, Bing Liu, Chen Xing

Scale AI

arXiv:2510.02663v1 [cs.LG] 3 Oct 2025

ABSTRACT

As students increasingly adopt large language models (LLMs) as learning aids, it is crucial to build models that are adept at handling the nuances of tutoring: they need to identify the core needs of students, be adaptive, provide personalized guidance, and be accurate. To this end, we introduce TUTORBENCH, a dataset and evaluation benchmark designed to rigorously evaluate the *core tutoring skills* of LLMs. The dataset comprises 1,490 samples curated by human experts, focused on high-school and AP-level curricula. The samples are drawn from three common tutoring tasks: (i) generating adaptive explanations tailored to a student's confusion, (ii) providing actionable feedback on a student's work, and (iii) promoting active learning through effective hint generation. To account for the inherent complexity of tutoring, samples are accompanied by sample-specific rubrics which are used to judge model responses during evaluation. TUTORBENCH uses a reliable and fine-grained automatic evaluation method that uses an LLM-judge and the sample-specific rubrics. We evaluate 16 frontier LLMs on TUTORBENCH and present a detailed analysis of their performance and behavior. Our results show that none of the frontier LLMs achieve a score of greater than 56%, showing a large room for improvement. We find that LLMs fall short in exhibiting the full range of tutoring skills needed to guide, diagnose, and support students effectively, with all the frontier models achieving less than a 60% pass rate on rubric criteria related to these skills. We also find that different model families exhibit varied strengths and limitations: the Claude models outperform others in supporting active learning, while they lag behind in the other two use cases. By releasing TUTORBENCH, we provide a comprehensive and unsaturated benchmark to guide the development of the next-generation of AI tutors.

1 INTRODUCTION

Large language models (LLMs) are rapidly transforming the way students learn and access educational support (Handa et al., 2025; Ammari et al., 2025; Geraghty & Goldstein, 2024; Scarlatos

Simulating human belief updating

LLMs struggle to simulate human belief updates in controlled environments

Sebastian Pohl[1,*], Harsh Mehta[1], Pranav Mambayil[1], Abdul Ghafoor[1], Franziska Lesigang[1], Yufang Hou[1,†], Christian Hilbe[1,†]

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arXiv:2607.28347v1 [cs.CL] 30 Jul 2026

Abstract

LLMs are increasingly deployed as proxies for human study participants in social science experiments, yet the fidelity of this practice has rarely been tested directly. We test whether six LLMs can simulate individual human belief updates, comparing LLM outputs 1-to-1 against ground truth data from 391 UK participants on Prolific, who updated their stances on three discussion topics after reading Reddit comments. Each participant was simulated by an LLM conditioned on a persona derived from their demographic and personality trait data. We find that some LLMs (Qwen3-32B and GPT-5-Mini) can match the human post-stance distribution, but only when given participants' actual initial stances. All six models fail to simulate initial stances themselves and to produce faithful belief updates from self-generated stances. Three systematic biases emerge across all models: overrepresentation of neutral positions, more frequent but smaller belief shifts than humans, and a failure to rank comments by convincingness. Demographic and personality trait personas had no consistent effect on fidelity. LLM simulations of human belief dynamics are only reliable when grounded in realistic starting conditions, that current multi-round social media simulations rarely

Building simulated classroom environments

Classroom Simulacra: Building Contextual Student Generative Agents in Online Education for Learning Behavioral Simulation

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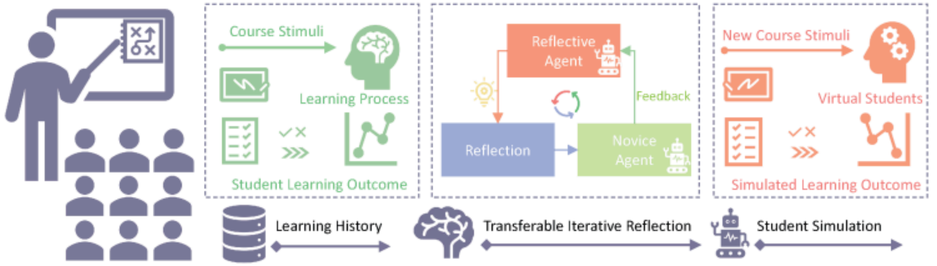


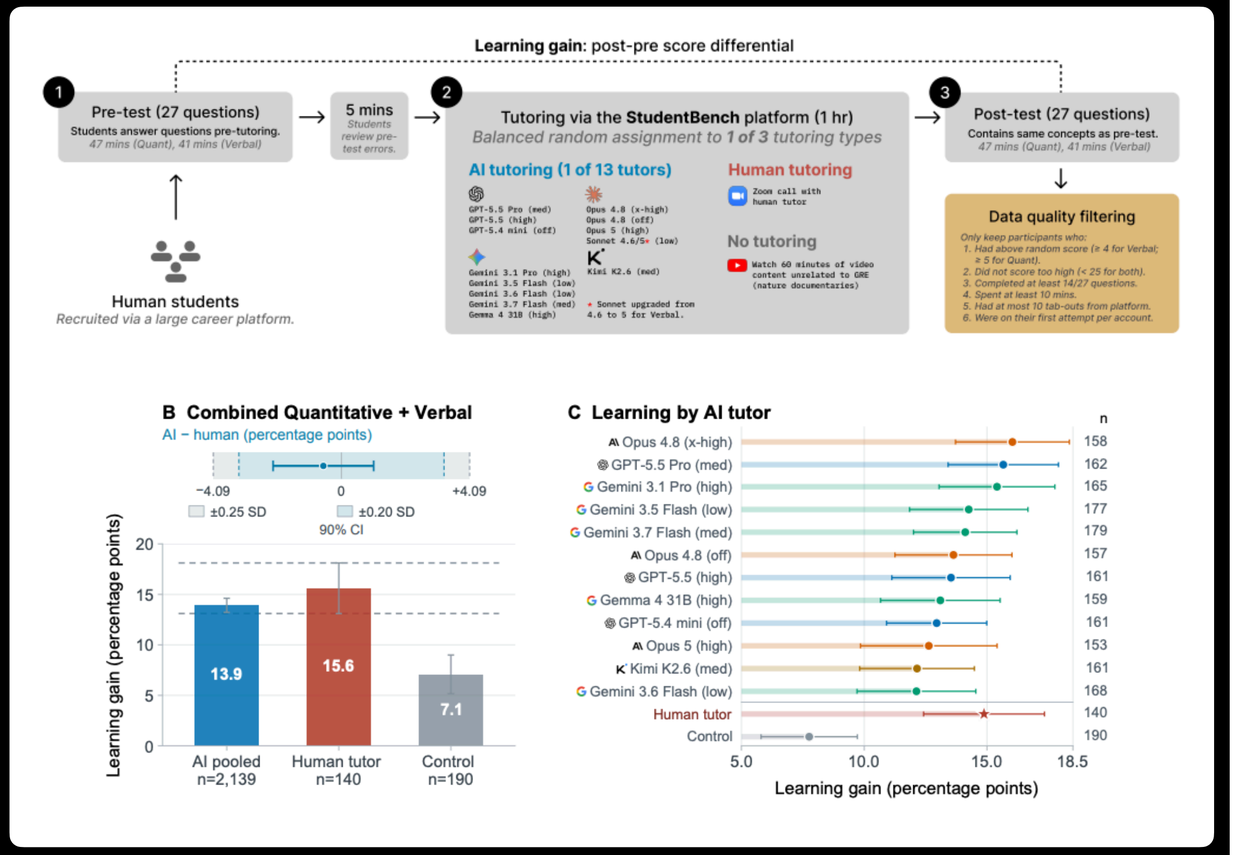
Figure 1: Overview of our Classroom Simulacra framework which uses a transferable iterative reflection method to effectively learn from students' history and capture how course materials modulate learning behaviors, so as to enable realistic student simulation.

Abstract

Student simulation supports educators to improve teaching by interacting with virtual students. However, most existing approaches ignore the modulation effects of course materials because of two

and finetuning-based large language models (LLMs) for simulating learning behaviors. Our comprehensive experiments show that TIR enables the LLMs to perform more accurate student simulation than classical deep learning models, even with limited demonstration

Evaluating real learning outcomes of agents



arXiv:2609.28470v2 [cs.AI] 30 Sep 2026

StudentBench: AI and human tutoring yield equivalent GRE learning gains

Curtis Northcutt, Inaara Hasmani, Kevin Feng, Trevor Khangi, Andreas Plesner, Jonas Mueller

Artificial intelligence offers an unprecedented opportunity to augment human capabilities, yet progress at the frontier has focused primarily on advancing model capabilities. We introduce StudentBench, a suite of AI teaching evaluations and a public platform that enables large-scale data collection with over 175,000 student–AI messages to study whether large language models (LLMs) produce learning gains equivalent to human tutoring. Using StudentBench, we measured learning gains on Quantitative and Verbal GRE questions across 2,383 human participants receiving AI tutoring, human tutoring, or no tutoring. We establish that AI tutoring is statistically equivalent to expert human tutoring for GRE learning gains (p=.015), and in five of the seven GRE domains, the best performing AI tutor surpassed the human tutor, on average. In a second study, expert human tutors compared LLM-generated lesson plans and practice problems through 2,028 pairwise rubric evaluations. Together, the two studies clearly separate AI tutors across: (1) lesson planning, (2) practice-problem creation, (3) conversational pedagogy, (4) cost, and (5) engagement. Surprisingly, one AI tutor achieved learning gains equivalent to human tutoring (p=.044) at 918× lower cost ($0.0052 for AI versus $4.81 for human, per percentage point gained). For Quantitative GRE sessions, faster AI replies correlated with more student messages, more messages with more correct practice, and more correct practice with larger learning gains (all $p \leq .002$). To support future research, we open-source the de-identified data collected in our studies.

Date: September 2026

Handshake AI

1 The pursuit of human-level tutoring

A singularity in intelligence can occur when an AI technology recursively self-improves, such that there is an irreversible split in intellectual ability between those who embrace the technology and those who do not (Good, 1965). While work in artificial intelligence has focused on the historical moment of recursive self-improvement (RSI) for AI itself (Vinge, 1993; Zhang et al., 2026a,b), we examine a parallel singularity for human intelligence.

Humans augmented by teaching technologies may recursively self-improve faster than those who do not use them. We use the term *recursive human self-improvement* (RHSI) for this process: better teaching technologies help people learn, and greater human capabilities can in turn support better use and development of those technologies (Engelbart, 1962). Establishing RHSI requires three conditions: first, the technology improves human capabilities; second, an improved human is better at using the technology; and third, better use leads to

Sotopia: Social Interaction Environment

Strong benchmark for negotiation, coordination, competition, social intelligence

sotopia-lab/

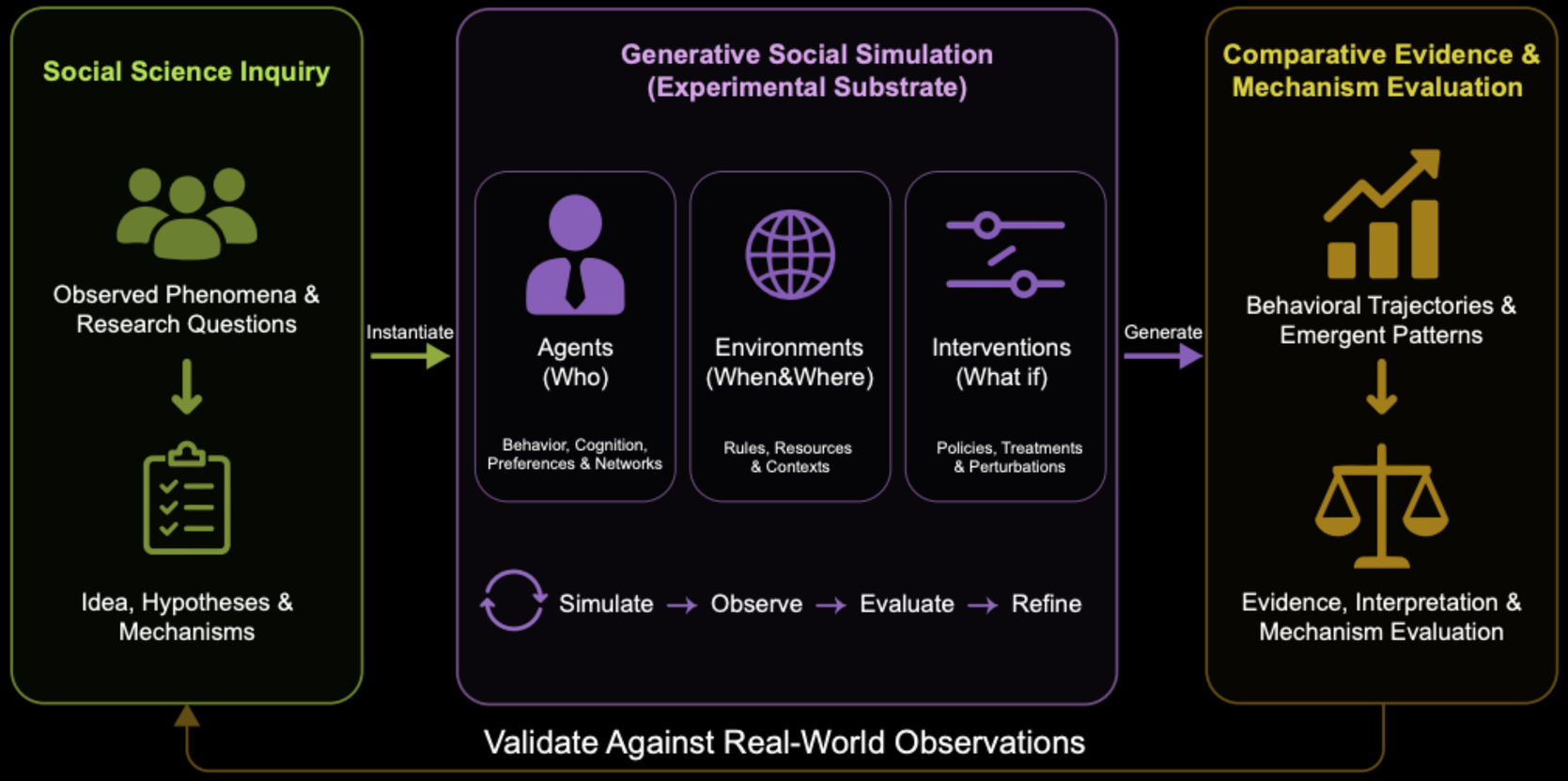
Sotopia

an Open-ended Social Learning Environment

```
pip install sotopia
```

Simulate Community: How can an agent influence well-being in a community?

AgentSociety-2 is typically used for conducting social science research



AgentSociety [2]

How do we evaluate human outcomes?

How do we evaluate human cognitive outcomes?

A. **Evaluate agent behavior / capability.** E.g. does the tutor agent exhibit behaviors that learning-science theory says are desirable? (MathTutorBench, MRBench, ImpactBench, RealToxicityPrompts, SyncEval)

B. **Evaluate simulated human states. E.g.** EduClaw-Bench, “LLMs struggle to simulate human belief updates in controlled environments” (Pohl et al. 2026)

C. Measure real human states. Real humans are exposed to an intervention (agent system) and their outcomes are measured via cognitive tests, behaviors or surveys. E.g. randomized controlled trials, pre/post designs, delayed retention tests, behavioral measurements.

Simulations are not always accurate

Ran customer service agent tasks similar to **τ²-bench** with LLM users and 451 real humans. LLM users were more cooperative, less ambiguous/ frustrated, and made agents look better than real humans did. They introduce the User-Sim Index (USI) to measure similarity to real interactive behavior.

Published as a conference paper at COLM 2026

Mind the Sim2Real Gap in User Simulation for Agentic Tasks

Xuhui Zhou* **Weiwei Sun*** **Qianou Ma** **Yiqing Xie** **Jiarui Liu** **Weihua Du**
Sean Welleck **Yiming Yang** **Graham Neubig** **Tongshuang Wu** **Maarten Sap**
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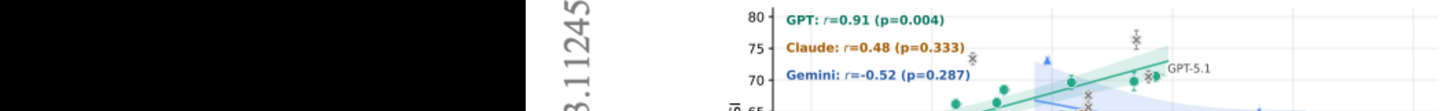
arXiv:2603.11245v2 [cs.AI] 31 Jul 2026

Abstract

As NLP evaluation moves from static benchmarks to multi-turn interactive settings, LLM-based simulators have become widely used as user proxies, serving two roles: generating user turns and providing evaluation signals. Yet, these simulations are frequently assumed to be faithful to real human behaviors, often without rigorous verification. We formalize the *Sim2Real gap* in user simulation and present the systematic study running the full τ-bench protocol with real humans (451 participants, 165 tasks), benchmarking 31 LLM simulators across proprietary, open-source, and specialized families using the *User-Sim Index (USI)*, a metric we introduce to quantify how well LLM simulators resemble real user interactive behaviors and feedback. Behaviorally, LLM simulators are excessively cooperative, stylistically uniform, and lack realistic frustration or ambiguity, creating an "easy mode" that inflates agent success rates above the human baseline. In evaluations, real humans provide nuanced judgments across eight quality dimensions while simulated users produce uniformly more positive feedback; rule-based rewards are failing to capture rich feedback signals generated by human users. Overall, higher general model capability does not necessarily yield more faithful user simulation. These findings highlight the importance of human validation when using LLM-based user simulators in the agent development cycle and motivate improved models for user simulation.

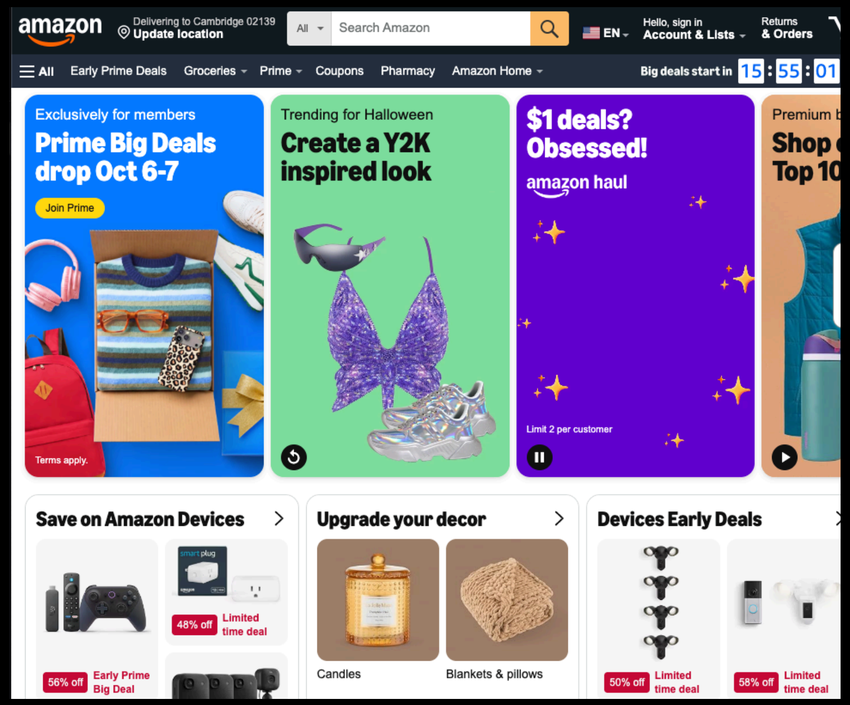


How can we validate a simulated environment?

- **Perceived realism.** Do humans/experts judge it as realistic? E.g. accuracy ratings.
- **Real-world data alignment.** Does simulated behavior match existing real-world data? E.g. do states and transition dynamics between states match real-world states and transition dynamics?
- **Real-world outcome validity.** Do results from simulation reproduce in reality? E.g. do the effects caused by an agent in the simulation carry over to the real world.
- **(Bonus) Robustness across simulators.** Do conclusions hold if we change the simulator/LLM-model to another?

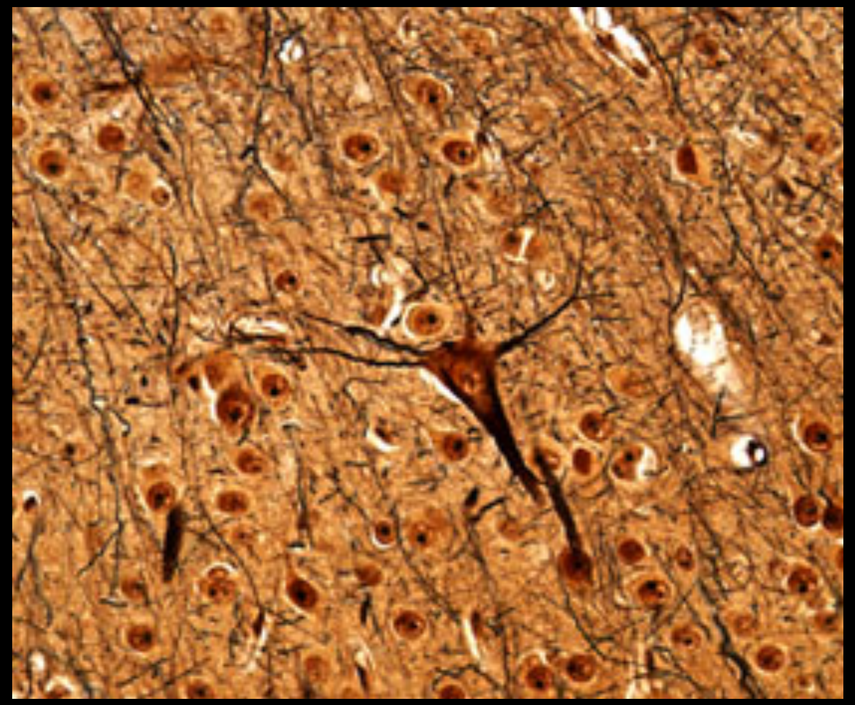
Examples of validations

- **Real-world data alignment**
  - GreenLight-Gym
    Compared simulated greenhouse climate/crop dynamics against measured greenhouse data.
  - Simulated Students in Tutor Dialogues (Scarlatos et al.,2026)
    Compared simulations dialogues against real tutoring dialogues using separate linguistic, behavioral/dialogue act, and cognitive metrics like a knowledge-tracing model.
  - Simulated people in decision making scenarios (Park et al. 2025),
    “Generative Agent Simulations of 1,000 People.” Compared simulations of real people to their decision making game outcomes.
- **Real-world outcome validity**
  - Simulated Customers (Zhou et al., 2026)
    Ran tasks with LLM users and 451 real humans and compared benchmark accuracy.


amazon
Delivering to Cambridge 02139
Update location
All
Search Amazon
EN
Hello, sign in
Account & Lists
Returns
& Orders
All
Early Prime Deals
Groceries
Prime
Coupons
Pharmacy
Amazon Home
Big deals start in 15 : 55 : 01
Exclusively for members
Prime Big Deals drop Oct 6-7
Join Prime
Terms apply.
Trending for Halloween
Create a Y2K inspired look
$1 deals? Obsessed!
amazon haul
Limit 2 per customer
Premium b
Shop
Top 10
Save on Amazon Devices
48% off Limited time deal
56% off Early Prime Big Deal
Upgrade your decor
Candles
Blankets & pillows
Devices Early Deals
50% off Limited time deal
58% off Limited time deal

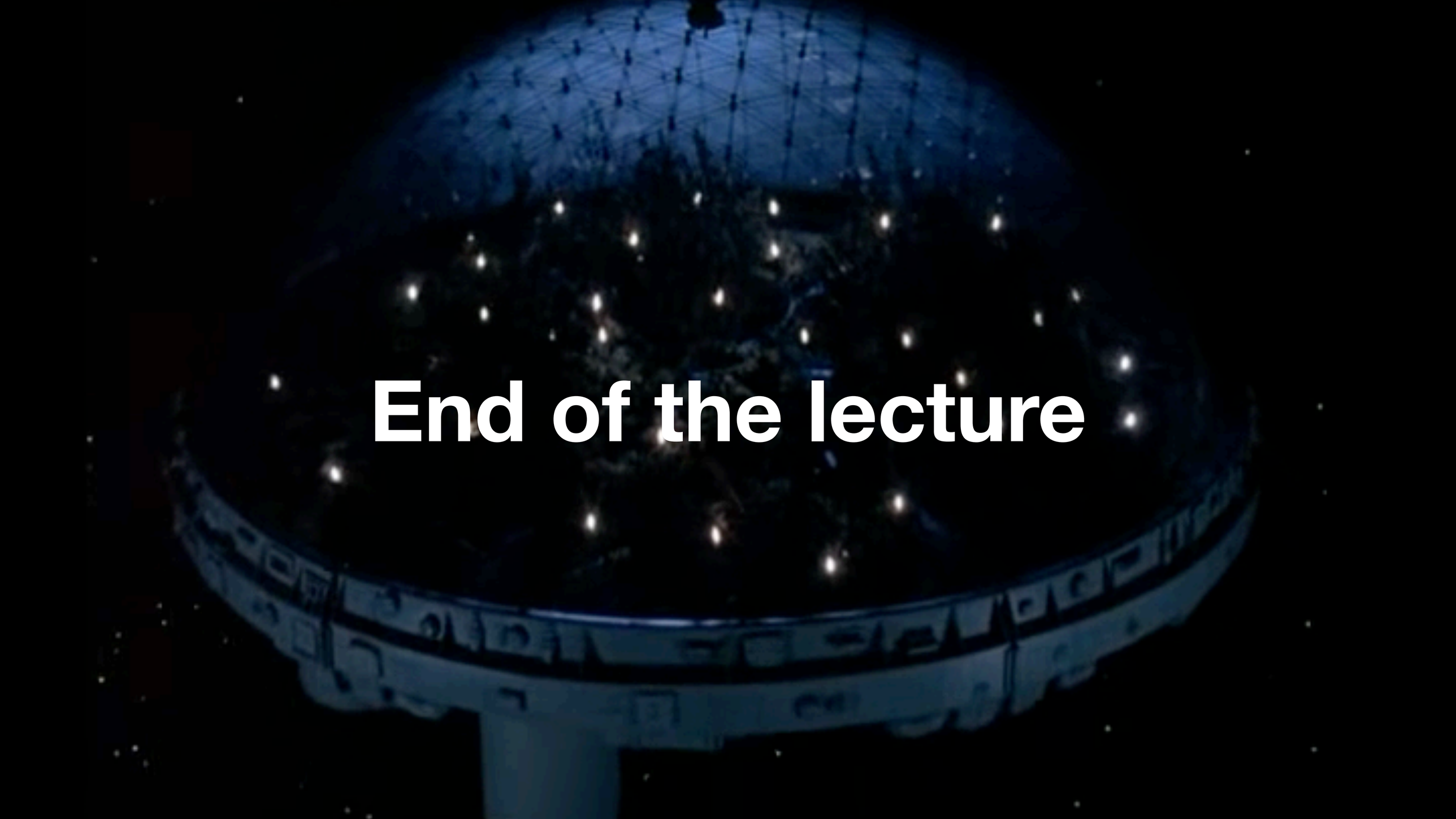
DRINKS
TECH
SNACKS
WELLNESS
TOUCH ANYWHERE TO START!

Credit: Hiroshi Ishii



Provocation: Can we benchmark agents in real environments?

real vending machine
real cats (under supervision)
real plants
real students, etc.

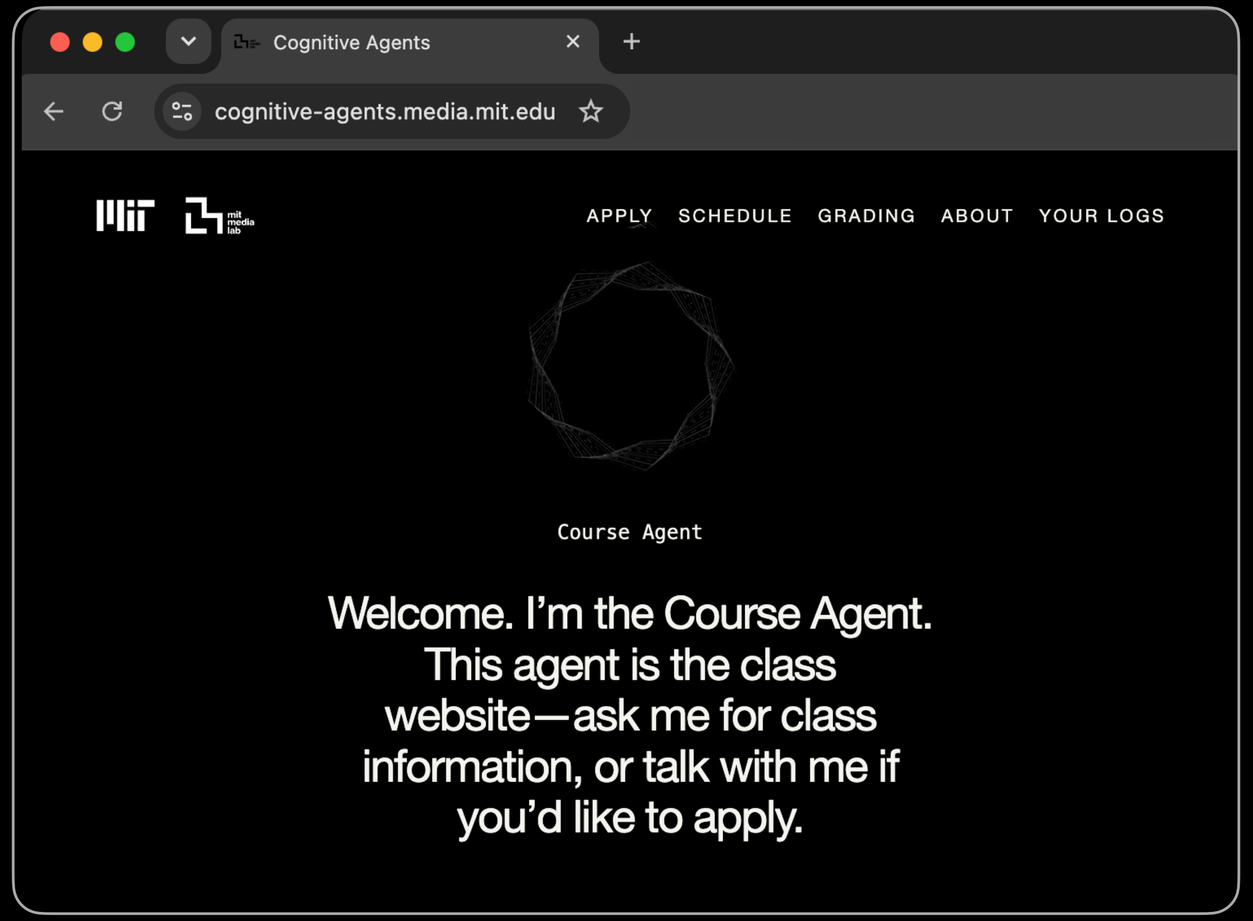

End of the lecture

Lets socialize!

- which environments?
- where do they fall on the

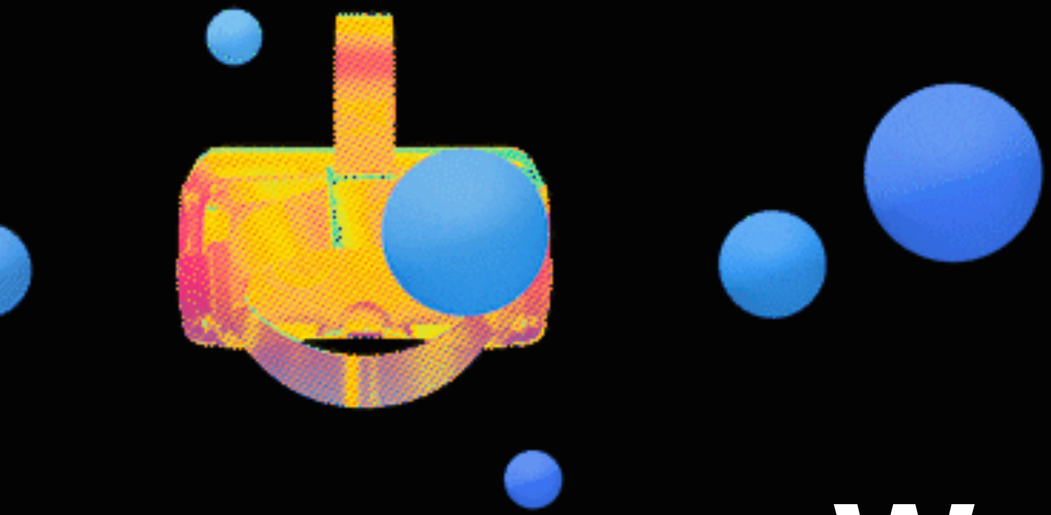
  (1) fully observable vs. partially observable
  (2) single-agent vs. multi-agent
  (3) deterministic vs. stochastic states
  (3) episodic vs. sequential
  (4) static vs. dynamic
  (5) discrete vs. continuous
  (6) known vs. unknown

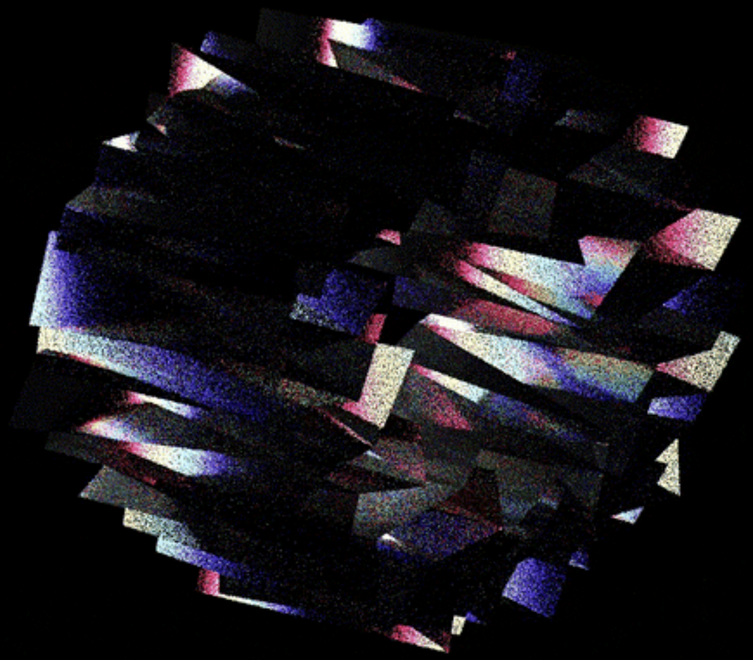
- how to validate their accuracy?
- how to randomly initialize them?
- which tasks? And scored how?

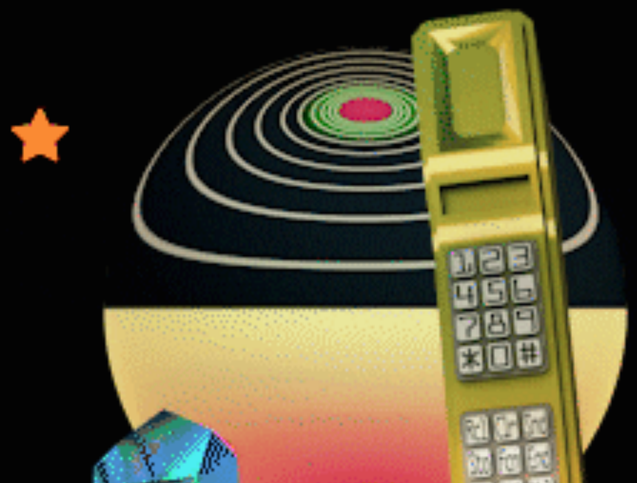


Let’s start the tutorial

Week 4 build



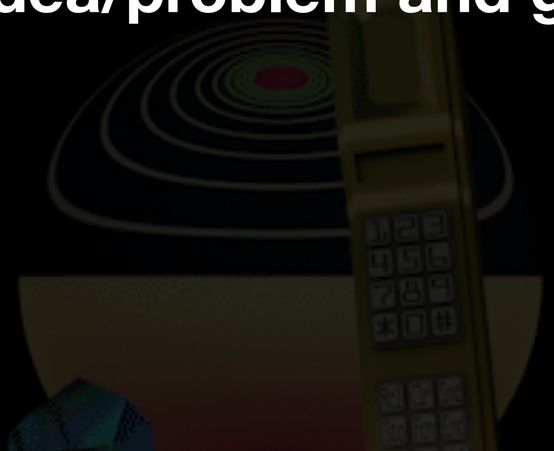




Week 4 build

. build one realistic environment with a scoring and evaluate a basic agent running objectives.

. submit a final idea/problem and group for the group project



Open Labs

Mon and Thur
3-4 PM

No class next week!

Thank you and looking forward to next week!